

Vons: Compact Decision Models for Frontier-Agent Workflows

Technical Report v1.1

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Abstract

Vons is a compact decision model for frontier-agent workflows: a frontier model plans and proposes candidates, Vons scores the current state against those candidates and returns a structured decision with an abstention option, and the host alone validates, authorizes and executes. This Technical Report is the engineering record behind the companion paper. It is bound to one hash-verified evidence packet (`publication-evidence-v2.zip`) and describes the architecture and the KASI host boundary, the implementation that exists, the synthetic-pilot results under `vons.metrics/v2`, the shared ONNX bundle and its CPU parity, a four-cell CPU padding decomposition, historical browser timings, the first external evaluation on the original Mind2Web test splits, the local-model suitability screen, provenance and reproduction commands, a failure taxonomy, and the release gates that remain open. Measured findings, each bound to a field-level ledger row (Appendix A): a shared encoder plus both decision heads and required model assets totals 46,001,618 bytes (43.8706 MiB) against the 67,108,864-byte (64 MiB) design target, whereas the two original separate bundles sum to 91,753,063 bytes and do not jointly meet it; 80 finite CPU graph-partition comparisons pass at $\text{atol} = \text{rtol} = 10^{-5}$ with maximum absolute difference 0.0. On the 200-row synthetic test split the Direct head reaches answerable accuracy 1.0 at coverage 1.0 with selective risk 0.09 (covered unanswerable rows count as errors) and ECE 0.115–0.288; because all 182 answerable rows carry the same label `call_tool`, a constant predictor also reaches 1.0, so this accuracy carries no discrimination evidence. The four-step Diffusion head reaches answerable accuracy 0.011–0.060 at coverage 0.11–0.29; the two heads are reported separately and are not ranked. One calibration fit on a separate 200-row calibration split yields coverage 0.91 and selective risk 0.0 on the test split with the temperature at its search lower bound. On Mind2Web (official revision, three held-out splits, $k = 5/10/20/32$) there were 37,512 selection attempts: 31,769 strict 512-token overflows, 5,743 confidence abstentions and 0 accepted selections; candidate recall rises with k to about 0.29–0.32 task-macro at $k = 32$, conditional selection accuracy is 0.0 and complete-case accuracy is undefined. Historical browser measurements of the Direct full graph (500 timed samples per requested provider over five page/session starts) give total-request p50 of 2207 ms (requested WASM) and 91 ms (requested WebGPU); provider execution is unverified, page starts are not independent process launches, and process/GPU memory is null. No completed browser numerical parity matrix exists; the stalled attempt is not a pass. The Jev comparison is not run. A second, separate body of evidence is added in this version: a post-freeze remediation round dated 2026-09-25, supplied to the drafting session as a Codex-verified handoff of file paths, SHA-256 digests and values, not contained in the frozen packet and not independently rerun or accessed by the drafting assistant (Section 8; label `POST-FREEZE (HANDOFF)`). Stated exactly as handed off: dynamic per-request padding in Python and TypeScript with a fresh local CPU shape benchmark (80 samples per cell; Direct 32 slots $\times$ 64 tokens inference + readback p50 19.750145 ms, passing the < 20 ms gate narrowly) and a read-only re-verification of the frozen shared-v1 partition (80 samples, maximum absolute differences 0.0); a balanced four-class synthetic set (pilot-v2, 2,000 rows) with vector-scaling calibration v3 whose per-class temperatures are strictly interior (calibrated Direct test macro-F1 1.0 against a

constant-majority 0.101732, coverage 0.92, selective risk 0.0, ECE 0.036557); a token-conditioned Diffusion head reaching answerable accuracy and macro-F1 1.0 on that synthetic test (coverage 0.92, Brier 0.0136773); and a label-free Mind2Web context compression under which all 37,512 attempts are accepted with 0 overflows and a conditional selection accuracy of 0.0581668 under an explicitly uncalibrated acceptance gate. The frozen v1 results remain the historical baseline; the v2 evidence is synthetic-only for quality, local-CPU-only for latency, graph-partition-only for parity and released-candidate action-target selection (not browser task success) for Mind2Web. Release gates were not re-audited and the release assessment (hold) is unchanged. Nothing here is a product validation claim.

Licence and status. The report text and its original figures are licensed under CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>). The licence does not cover the Vons software, quoted source excerpts, third-party assets, raw benchmark data or private verification records (Section 13). Report type: pilot / negative-result engineering record accompanying the companion paper. Version 1.1 (September 2026) keeps the frozen v1.0.1 record unchanged and adds, in a separately labelled section, post-freeze remediation evidence dated 2026-09-25 that was supplied as a Codex-verified handoff and was neither rerun nor accessed by the drafting assistant (Section 1, Section 8). Every research release gate listed in Section 12 remains open exactly as recorded at the evidence freeze; no gate was re-audited for this version, and this version does not announce publication readiness. arXiv submission 8127259 is on hold, and this version does not itself update or publish it. Every frozen measurement was read from one frozen evidence packet and was not re-run for this version.

1 Scope, Evidence Basis and Reading Conventions

Authoritative snapshot. Only the frozen archive `publication-evidence-v2.zip` (SHA-256 `f2fb06c2275a60eec6c5b89938123da599b9979bb10d7e65a5667f6ca69152c4`) is used. Its manifest `packet.json` (SHA-256 `7cc5b19cd0e7691919d632d431ff1101d5a5b26a7c11c104c9fa83c132cd5318`; schema `vons.publication-evidence/v2`; frozen 2026-09-24T08:41:25.776377+00:00) lists 131 files; the archive has 133 members, the two additional members being `packet.json` itself and `input-spec.json`, which is covered by the manifest’s `spec_sha256` field. All 131 listed files and all 258 supplied claim rows were verified in the drafting session (131/131 files matched by SHA-256 and byte count; 258/258 claims matched their source field). The verification record accompanies this report as a private file (Section 9). Earlier drafts and registers are preserved and are superseded where they conflict (Section 14).

Code revision (frozen capture state). `packet.json` records `git_revision = null` with `git_revision_reason` “workspace has no Git repository” (DOCUMENTED STATUS). This describes the **frozen capture state at the 2026-09-24 freeze**: the workspace from which the packet was frozen was not a Git checkout, so provenance rests on the explicit per-file SHA-256 digests in the manifest and on the bundle-manifest digests for model weights. A repository has since been created by the author (Section 13); no commit of that repository can be bound retroactively to any measurement in the packet.

Post-freeze evidence (new in v1.1). A second body of evidence, dated 2026-09-25, is reported in Section 8 and nowhere else in the results sections. It reached the drafting session as a handoff message from the integrator (Codex) listing repository file paths, their SHA-256 digests and the values read from them; the primary index is `reports/goal-v2/evidence-summary.json` (SHA-256 `39f53771bbf4ffe454dbac09412baf0f371804e6442bb197eaf9879afbfc640a`). **None of these paths is a member of the frozen packet `publication-evidence-v2.zip`**, whose digest and 133-member inventory are unchanged (one handoff-stated digest coincides with a frozen member; Section 8.1), and none was opened, hashed, rerun or otherwise accessed in the drafting session. Every value from this handoff carries the label `POST-FREEZE (HANDOFF)` and the ledger label “POST-FREEZE MEASURED (Codex-verified handoff; not independently rerun by Claude Science)” (Appendix A); in prose it is described as post-freeze evidence, 2026-09-25 (Codex-verified handoff; not independently rerun or accessed by Claude Science).

Post-freeze values are never merged with frozen values in one table cell, one range or one ratio; where a post-freeze quantity has a frozen counterpart, the two are shown in separate columns with their differing sample sizes and settings stated.

Input verification is not re-execution. Verifying digests establishes that the reports in the packet are the reports the integrator froze; it does not reproduce their measurements. Nothing in this report was retrained, re-inferred, re-benchmarked or re-run in a browser for any version of the report. The Mind2Web raw archive, normalized rows, per-row predictions and all model weights are deliberately absent from the author packet; their digests and aggregate results are present (Section 6, Section 9).

Labels. Every quantity carries one of the following labels. **MEASURED:** a value transcribed from a named field of a hash-verified file (ledger operation identity). **MEASURED (HISTORICAL):** measured on an earlier configuration that is not the current software. **DERIVED:** an arithmetic combination of measured fields, with the formula recorded in the ledger. **DESIGN TARGET:** an intended property, never an outcome. **DOCUMENTED STATUS:** an integrator-recorded implementation state, not independently executed here. **PLANNED:** intended work, not implemented or not run. **UNAVAILABLE:** not measured, or `null` in the source. **AUTHOR DECISION:** a licensing or status choice made after the evidence was frozen. **POST-FREEZE (HANDOFF):** a value stated in the 2026-09-25 Codex-verified handoff of post-freeze evidence; not in the frozen packet, and not independently rerun or accessed by the drafting assistant (Section 8). Values in tables are rounded to the precision shown (at most four decimals; milliseconds to three); the full-precision values are in the named packet fields. Every quantitative statement is bound to a row of the claim ledger that accompanies this report; the body text no longer carries ledger identifiers, which are collected per section and table in Appendix A.

What this report does not claim. Product readiness; external-task quality (frozen Mind2Web selection is a negative result; the post-freeze conditional selection accuracy is released-candidate action-target selection under an uncalibrated gate, not browser task success); browser-deployment performance of the current SDK or shared bundle; peak memory; general Diffusion inferiority or superiority; general calibration; verified KASI boundary behaviour on audit logs; a Jev comparison; an adopted open-source licence; independent verification of any post-freeze value; closure of any release gate by the post-freeze evidence.

2 Architecture and the Host/KASI Boundary

Decision contract. A Vons request carries a structured state, a bounded question and a candidate list; the model returns a choice with a probability distribution and an answerability/abstention signal. The documented SDK limits are 512 input tokens, 8 questions per request and 32 candidates (**DOCUMENTED STATUS**). Three things about Score questions must be kept apart. (a) The generic contract *schema* admits Score questions with a 2–10 item rubric (`vons/contract.py` validator; `tools/verify_browser_parity.py` accepts the type for request validation only). (b) *Runtime support:* the candidate-selection ONNX head in the browser SDK **rejects** Score questions (`sdk/typescript/src/onnx.ts` throws “score questions are not supported by this candidate-selection ONNX head”), and no Python inference backend in the frozen snapshot implements Score inference; no Score inference is implemented or measured anywhere in the packet (**DOCUMENTED STATUS**). (c) An expected-ordinal score and a boolean $P(\text{true})$ mapping are **PLANNED** evaluation-adaptor behaviour, not current contract outputs. Model-written confidence is never treated as a calibrated probability; calibration is fitted only on a disjoint calibration split (Section 4.4). The Python validator estimates the token budget from characters whereas the TypeScript SDK uses a tokenizer budget, so the two implementations do not yet form one fully harmonized public contract.

KASI is a compatibility layer, separate from the general contract. KASI keeps the five-action public contract—`call`, `clarify`, `confirm`, `refuse`, `respond`—on top of Vons’ general choice and abstention

Table 1. Division of authority in the KASI host adapter (DOCUMENTED STATUS: implemented in the frozen source and covered by unit tests; not measured on live workflows).

Decision	Made by	Rule in the frozen source
Propose a call, clarification, refusal or response	Model	Proposal only; never executed by the model
Whether a tool may be used	Host	Unregistered tools are refused
Risk of a tool	Host registry	Advisory proposal risk cannot lower registered risk; missing or unknown risk fails closed
Whether confirmation is needed	Host	confirm is derived from host policy, never asserted by the model
Consent	User, via host	Bound to the complete call including arguments (KASICall.fingerprint() / canonical {name, arguments} key)
Execution	Host	The host is the only execution boundary

machinery. The model may *propose* ordinary calls, clarification, refusal or a response. It cannot grant consent and never executes. The host wrapper requires a host-owned registry of tool names and risk levels, validates tools and arguments, applies confidence thresholds, and **derives confirm from that policy**. The proposal’s risk field is advisory only: it can never lower the registered host risk, and missing or unknown risk fails closed. Consent is bound to the complete call including its arguments through a `KASICall.fingerprint()` SHA-256 (Python) or the canonical {name, arguments} key (TypeScript); a bare tool name in `confirmed_tools` cannot authorize a high-risk call. Unregistered tools are refused. The mapping is one-way: KASI calls can be represented as Vons choices, but arbitrary Vons choices do not become tool calls without the host adapter. Risk metadata and the confirmed-call set are therefore **KASI-side host inputs**, not fields of the general decision request. Table 1 summarizes the division of authority.

Verification status. The Python adapter (`vons/kasi.py`) and the TypeScript policy (`sdk/typescript/src/index.ts`) are in the snapshot and are covered by the integrator’s unit tests (241 Python tests with 50 subtests; 10 SDK tests; DOCUMENTED STATUS). The post-freeze handoff states a later implementer check of 291 Python tests plus 50 subtests and 16 TypeScript tests (POST-FREEZE (HANDOFF); Section 8.5); the adapter and boundary rules described in this section are unchanged by the handoff. No adapter audit log from a real workflow exists in the packet, so boundary metrics—confirm agreement, under- and over-confirmation, tool validity, invariant violations—remain UNAVAILABLE. The boundary is a design requirement whose unit-level behaviour is tested and whose end-to-end behaviour is unverified.

3 Implementation Status

3.1 Components present in the snapshot

Table 2 lists what exists in the frozen source and where the packet evidences it. Every entry is a DOCUMENTED STATUS; none was re-executed for this report.

3.2 Gate status

Table 3 transcribes the integrator’s gate table from `docs/COMPLETION_PLAN.md`. A status entry is not completion evidence. The plan text refers to an intermediate packet digest; this report binds to the v2 packet named in Section 1.

Post-freeze evidence and the gate table. The gate table is the frozen record and was **not re-audited** for this version. The post-freeze handoff (Section 8) reports work that bears on G3 (the “dynamic redesign” is reported as implemented in Python and TypeScript with a fresh local CPU benchmark) and on G4 (a complete v2 Mind2Web matrix with a compressed input), and it reports no browser measurement, so G1, G2 and the browser items are untouched. Whether any recorded status changes is a decision for a future gate re-audit; this version changes no entry (POST-FREEZE (HANDOFF)).

Table 2. Components present in the frozen snapshot (DOCUMENTED STATUS); the last row is the post-freeze handoff statement (POST-FREEZE (HANDOFF)) and describes a later code state that is not in the snapshot.

Area	What exists	Evidence in packet
Shared frozen encoder / tokenizer	google/bert_uncased_L-4_H-256_A-4 at revision 387825ce42db; 512 tokens...	reports/pilot-direct-test-calibrated-v2.json config
Direct head	One-pass candidate scoring; exported ONNX head and full graph	artifacts/pilot/model-manifest.json; browser-direct bundle manifest
Diffusion head	Conditional score denoising; deterministic DDIM, 4 steps inside one graph; 32 fixed slots; host-supplied initial_noise	artifacts/pilot-diffusion/model-manifest.json; shared-v1 heads
Shared-v1 bundle	One encoder + two heads + tokenizer/config/calibration; transformation recorded as “partition existing graphs; no training or parameter changes”	artifacts/shared-v1/manifest.json
Python package vons/	Contract, data, models, training, evaluation (metrics/v2), calibration (calibration/v2), bundle manifest, KASI adapter, Mind2Web importer/evaluator, Ollama benchmark, research packet	13 modules and 19 Python test files under tests/
TypeScript SDK	Contract validation, KASI host policy, explicit WASM/WebGPU capability selection; optional ./onnx runtime that verifies the manifest, loads the tokenizer and .onnx.data, and runs both pilot bundles through pinned ONNX Runtime Web 1.30.0 [1]	sdk/typescript/src/{index,onnx}.ts, demo harnesses, 3 TypeScript test files; 10 SDK tests
Tooling	19 scripts under tools/ (Section 9.3)	Source snapshot, hashed
Integrator checks (2026-09-24)	.venv/bin/pytest -q: 241 passed, 50 subtests; ruff exit 0; SDK build exit 0	reports/orchestration-checkpoint-v2.json; not re-executed here
Implementer checks (post-freeze handoff, 2026-09-25)	291 Python tests passed plus 50 subtests; Ruff clean on all changed Python; 16 TypeScript tests passed; benchmark bundle build passed; git diff check passed; no commit or push	POST-FREEZE (HANDOFF): stated in the handoff; not in the packet; not re-executed here (Section 8.5)

3.3 Not implemented or not measured

The following are absent from the packet and are UNAVAILABLE or PLANNED: a shared-v1 browser loader; browser Score requests (explicitly unsupported); browser calibration parity; 2- and 8-step Diffusion browser paths; an additional seed matrix in the browser; lifecycle (cancel/dispose/re-initialize) and offline validation; matched padding cells in the browser; any Diffusion browser timing; browser process-

Table 3. Release-gate status recorded at the evidence freeze (DOCUMENTED STATUS; not re-audited).

Gate	Recorded status
G0 bundle manifest	Implemented and verified
G1 real WASM inference	Implemented; parity harness ready; full execution open
G2 browser matrix	Direct WASM/WebGPU controlled cells measured historically; Diffusion cells, fallback browser and lower-resource device open
G3 padding	CPU padding decomposition measured; dynamic redesign optional/open
G4 Mind2Web	Evaluated under evaluation-only authorization; negative result
G5 attribution	Attribution prepared; teacher inclusion open
G6 Jev comparison	Protocol frozen; comparison open
G7 public release	On hold
G8 licence	Proposal only at freeze (see Section 13 for the author decision made afterwards)
G9 Chrome-extension channel	Proposal only

level cold start; process or GPU memory; a complete browser numerical parity matrix; the Jev comparison; and a fully harmonized public character-budget contract between the Python estimator and the TypeScript tokenizer budget. This list describes the frozen snapshot. The post-freeze handoff states that dynamic per-request padding now exists in both the Python package and the TypeScript SDK and that a token-conditioned Diffusion head was exported with dynamic axes (POST-FREEZE (HANDOFF); Section 8.1, Section 8.3); the browser-side items above (loader, browser parity, Diffusion browser timing, memory, cold start) are not addressed by the handoff and remain UNAVAILABLE.

4 Synthetic Pilot

4.1 Data and metric definitions

The test split `data/generated/pilot-test.jsonl` has 200 rows (SHA-256 `e5fdd950a35593b97d9799435a8739177c1668d2ad7d30687546add0bc4de1bc`): 182 answerable and 18 unanswerable rows. The calibration split is a separate 200-row file, `pilot-calibration.jsonl` (SHA-256 `a986ab870f9d54d3905db1f3c71ea4e004922a5bdf43fe822e4c503d7dd3689`), drawn from the same 2,000-row generated set `pilot.jsonl` (SHA-256 `548090beedd52bc18d6a22de26a57efad81c11ddd125def94b63cfcaca9a8ee0`); the split sizes train 1,400 / development 200 / calibration 200 / test 200 are recorded in `artifacts/pilot/model-manifest.json` (MEASURED).

Label composition. Reading the test split directly: all 182 answerable rows are labelled `call_tool`, the 18 unanswerable rows have a null label, and every row has four options (DERIVED from the file). A constant `call_tool` predictor that never abstains therefore reaches answerable accuracy 1.0 and macro-F1 1.0 (a single class), so **synthetic accuracy carries no discrimination evidence**; only coverage, selective risk and the probability-based metrics distinguish behaviours on this split. The scope string in every quality report is “synthetic test split only; not Mind2Web or browser task success”. A balanced four-class replacement set (pilot-v2) is reported in the post-freeze section (POST-FREEZE (HANDOFF); Section 8.2); the frozen split above remains the basis of every table in this section.

Metric definitions. `vons.metrics/v2` corrects two v1 definitions: **selective risk** is errors over covered predictions, where a covered prediction on an unanswerable row is an error (v1 ignored those rows); **Brier** [2] is the standard multiclass score summed over options (range 0–2), whereas v1 divided each row by its option count. **NLL** and **Brier** are means over the *probability population*: answerable rows whose prediction has no error and carries a valid probability vector (`vons/evaluation.py`, `_probability_rows`, which applies `validate_distribution`); this population **includes abstained predictions** when a valid vector exists and therefore differs from the covered population used by ECE. Per row, $\text{NLL} = -\log(\max(p(\text{label}), 1e-12))$ and $\text{Brier} = \text{the sum over the row's options of } (p - 1[\text{option} = \text{label}])^2$, range 0–2; an empty population yields null. **ECE** [3] uses 10 equal-width confidence bins over covered predictions, with covered unanswerable rows counted incorrect. Because ECE is computed over all covered rows including unanswerable ones while answerable accuracy is not, the direction of miscalibration is **not** inferred from ECE alone in this report. The superseded v1 values are listed in Table 20.

Table 4. Direct head on the 200-row synthetic test split, `vons.metrics/v2` (MEASURED; rounded, full precision in `reports/quality-summary-v2.json`). Because the split is single-label (Section 4.1), answerable accuracy and macro-F1 equal the constant-predictor baseline and are uninformative about learning.

Seed	Ans. acc.	Macro-F1	Coverage	Sel. risk	ECE	Brier	NLL
7	1.0000	1.0000	1.000	0.0900	0.2163	0.0139	0.1053
17	1.0000	1.0000	1.000	0.0900	0.1154	0.0097	0.0871
27	1.0000	1.0000	1.000	0.0900	0.2881	0.0129	0.1013

Table 5. Diffusion head (four DDIM steps [4], 32 slots) on the same test split, `vons.metrics/v2` (MEASURED; rounded). Reported side by side with Table 4 and not ranked against it; no equal-effort tuning record exists.

Seed	Ans. acc.	Macro-F1	Coverage	Sel. risk	ECE	Brier	NLL
7	0.0275	0.0134	0.155	0.8387	0.4483	0.9318	1.8031
17	0.0604	0.0285	0.290	0.8103	0.4602	0.9370	1.8147
27	0.0110	0.0054	0.110	0.9091	0.5088	0.9368	1.8165

4.2 Direct head: three training seeds

Every seed is correct on all 182 answerable rows and covers all 200 rows—exactly the constant-label baseline of Section 4.1, so the 1.0 values establish only that the pipeline runs end to end. The 18 covered unanswerable rows give the uniform selective risk of 0.09 before calibration: the uncalibrated head never abstains. Cross-seed ECE spans 0.1154–0.2881; these are descriptive across saved seeds, not an independent test population. Post-freeze status: a calibrated Direct result on the balanced pilot-v2 test split is reported separately in Section 8.2 (POST-FREEZE (HANDOFF)).

4.3 Diffusion head: three training seeds (fixed four DDIM steps, 32 slots)

The fixed four-step head abstains on most rows (coverage 0.11–0.29) and is usually wrong when it answers (selective risk 0.810–0.909). Under the v2 definition (multiclass score summed over options, range 0–2) the Brier score is 0.932–0.937; no comparison with the v1 scale is made. The packet records the pilot’s design limitations—pooled-only conditioning, a weak terminal-noise schedule and decoder limitations—and states that the fixed four-step result **does not establish general Diffusion inferiority**. The two heads are reported side by side and are not ranked. Post-freeze status: a redesigned token-conditioned Diffusion head, trained and tested on the balanced pilot-v2 data, is reported separately in Section 8.3 (POST-FREEZE (HANDOFF)); it does not alter the values above, which remain the historical baseline for this head.

4.4 Calibration (Direct, seed 7; `vons.calibration/v2`)

Calibration first fits a temperature [5] on the calibration split and then fits an abstention threshold on temperature-scaled confidence for a design-target selective risk of 0.05, both on the separate calibration split (Table 6).

Reading. The pipeline runs in the intended order (temperature first, then threshold on temperature-scaled confidence), uses a disjoint calibration split, and on the test split abstains on exactly the 18 unanswerable rows, driving selective risk to 0 at coverage 0.91. The temperature fit saturated at its lower bound, so the temperature is a boundary value rather than an interior optimum, and ECE remains 0.223. This is pipeline evidence on one seed of one head on synthetic data; it supports no general calibration claim. Post-freeze status: `vons.calibration/v3` (vector scaling with interior per-class

Table 6. Calibration fit and calibrated test result for the Direct head, seed 7 (rounded to four decimals; full precision in `reports/pilot-direct-test-calibrated-v2.json`).

Quantity	Value	Label
Calibration split rows / temperature-fit rows / threshold-fit rows	200 / 182 / 200	MEASURED
Target selective risk	0.05	DESIGN TARGET
Fitted temperature (search bounds 0.05–20.0)	0.05, at lower bound	MEASURED
Abstention threshold; risk on calibration split	0.7558; 0.0	MEASURED
NLL before / after (calibration split)	0.1064 / 0.0	MEASURED
Answerability threshold	0.5	MEASURED
Test coverage / selective risk / answerable accuracy	0.91 / 0.0 / 1.0	MEASURED
Test ECE after calibration	0.2233	MEASURED
Abstained test rows	18 (= the unanswerable rows)	MEASURED

Table 7. Shared-v1 model-only bundle (MEASURED; runtime and notices excluded). Source: artifacts/shared-v1/manifest.json.

Asset	Role	Bytes	Label
encoder.onnx	encoder graph	553,978	MEASURED
encoder.onnx.data	encoder weights	44,423,168	MEASURED
direct.onnx	Direct graph	22,843	MEASURED
direct.onnx.data	Direct weights	132,096	MEASURED
diffusion.onnx	Diffusion graph	82,287	MEASURED
diffusion.onnx.data	Diffusion weights	74,752	MEASURED
tokenizer/tokenizer.json	tokenizer	711,396	MEASURED
tokenizer/tokenizer_config.json	tokenizer	379	MEASURED
calibration.json	calibration	233	MEASURED
config.json	configuration	486	MEASURED
Model-asset total		46,001,618 (43.8706 MiB)	MEASURED / DERIVED
Design target (64 MiB)		67,108,864	DESIGN TARGET
Size pass (manifest flag)		true	MEASURED
Runtime bytes		null (runtime not included)	UNAVAILABLE

temperatures) on pilot-v2 is reported separately in Section 8.2 (POST-FREEZE (HANDOFF)). Checkpoint digests are not carried in report configurations, so model identity is bound only by backend, model id, encoder and seed.

4.5 Harness head timings (not deployment latency)

The per-report `latency_p50_ms` values (0.049 ms Direct seed 7; 0.267 ms Diffusion seed 7) time the decision head only, with encoder embeddings precomputed; device and provider are not recorded. They are retained as historical harness fields (MEASURED (HISTORICAL)) and are not deployment or bundle latencies.

5 Deployment Artifacts

5.1 Shared-v1 bundle against the 64 MiB model-asset target

The shared bundle is a **partition of the existing graphs with no training or parameter change**; its manifest carries release status “research_candidate; parity and notices required before release”. The two original separate full bundles measure 45,853,779 bytes (Direct) and 45,899,284 bytes (Diffusion) of model assets (MEASURED (HISTORICAL)); each passes 64 MiB alone, but their sum, 91,753,063 bytes (DERIVED), does not—only the shared bundle makes both heads available within the target. The target governs model assets, not browser runtime, notices, RAM or first download; no licence or notice payload is yet part of shared-v1.

5.2 CPU graph-partition parity (not browser parity)

reports/shared-v1/parity.json: provider CPUExecutionProvider, ONNX Runtime 1.30.0, 2 intra-op threads, 20 cases giving 80 comparisons (20 Direct; 60 Diffusion over noise seeds 7/17/27), tolerance $\text{atol} = \text{rtol} = 10^{-5}$, overall pass true; maximum absolute score and answerability differences 0.0 (MEASURED). The file’s own scope is “graph partition only; no quality, browser, calibration or latency claim”. This establishes that partitioning preserved the graphs on CPU; it says nothing about browser execution, quality, calibration or latency. Post-freeze status: the frozen shared-v1 bundle was re-verified read-only (not modified) with the same 20 cases and 80 comparisons and maximum absolute differences 0.0 (POST-FREEZE (HANDOFF); Section 8.1); the scope remains graph-partition parity only.

Table 8. Four matched CPU cells (MEASURED; ms rounded to three decimals, full precision in reports/onnx-shape-decomposition.json). “Total” is the total request time per row; the derived comparisons in the text use the inference + readback p50 column.

Cell	Configuration	Samples	Total p50 (ms)	Total p95 (ms)	Infer.+readback p50 (ms)
A	Direct, live 4 slots $\times$ 64 tokens	240	2.586	2.711	2.416
B	Direct, 32 slots $\times$ 64 tokens	240	14.930	15.397	14.705
C	Direct, 32 slots $\times$ 512 tokens	240	219.344	232.942	219.078
D	Diffusion (4 DDIM steps), 32 slots $\times$ 512 tokens	240	220.622	232.749	220.367

5.3 CPU padding decomposition

reports/onnx-shape-decomposition.json: ONNX Runtime 1.30.0, platform macOS-26.6.2-arm64-arm-64bit, 3 session starts $\times$ 20 repeats $\times$ 4 rows = 240 samples per cell, 5 warm-up requests excluded per session, identical live texts, masks and noise across paired cells (MEASURED). This decomposition supersedes the earlier 20-row bundle timings of Tech Report v0.1, whose report files are not part of the v2 packet (Section 14). Table 8 and Figure 1 give the four matched cells.

Derived comparisons. The comparisons recorded in the file are differences of **inference + readback p50** (cell values 2.416, 14.705, 219.078 and 220.367 ms), not of the total-request column (DERIVED): candidate-slot padding within Direct (A $\rightarrow$ B, 4 $\rightarrow$ 32 slots at 64 tokens) adds 12.289 ms; sequence padding within Direct (B $\rightarrow$ C, 64 $\rightarrow$ 512 tokens at 32 slots) adds 204.373 ms; at the matched 32 $\times$ 512 shape the four-step Diffusion cell exceeds the Direct cell by 1.289 ms (D $-$ C). The matched difference is a system comparison at one shape on one CPU, still subject to architectural differences; it is **not** a method-only or DDIM-only cost and is not extrapolated to a dynamic export or to browsers. Post-

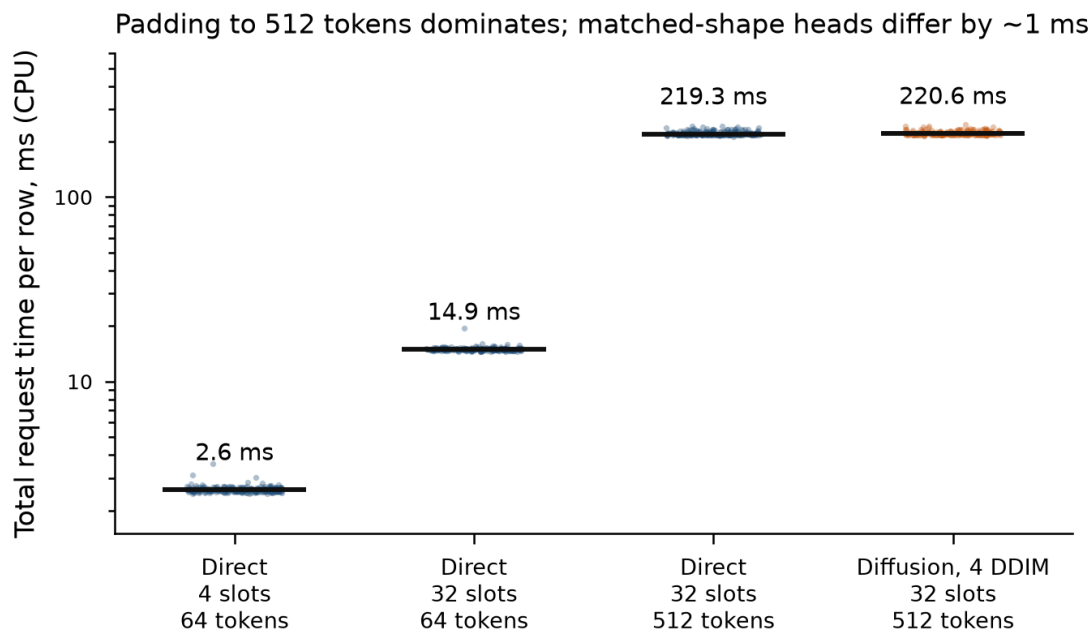


Figure 1. Total request time per row for the four matched CPU cells (log axis; 240 samples each; median marked). Historical full graphs on ONNX Runtime 1.30.0 CPU; not a browser measurement.

Table 9. Historical browser measurements of the Direct full graph (MEASURED (HISTORICAL); ms rounded to three decimals, full precision in the two *-trae-summary.json files). “Requested” means the execution provider was requested by the page, not observed.

Quantity	Requested WASM	Requested WebGPU
Timed samples (successful) / failed	500 / 0	500 / 0
Page/session starts	5	5
Total request p50 (ms)	2207.400	91.400
Total request p95 (ms)	2471.005	132.005
Inference + readback p50 (ms)	2207.000	91.000
Inference + readback p95 (ms)	2470.210	130.460
Tokenize p50 (ms)	0.300	0.200
Requested provider proven	false	false
Process-cold claimable	false	false
JS heap used before / after (bytes; not a peak)	11,757,347 / 11,811,939	57,884,604 / 57,884,604
Process RSS / GPU memory	null	null

freeze status: a fresh local CPU benchmark with dynamic per-request padding (80 samples per cell, different session settings) is reported separately in Section 8.1 (POST-FREEZE (HANDOFF)); the 240-sample cells above are not compared with it by ratio.

5.4 Historical browser measurements (Direct full graph; not the current SDK or shared bundle)

Both runs use the historical Direct full bundle whose manifest digest is 79c7f1e5e7f120e5d9f136fbfb3c6761e1783f73304f10a722b8a71a9966532b (the SHA-256 of artifacts/provenance/browser-direct/bundle-manifest-v1.json), 4 live candidates in 4 slots at 512 tokens, ONNX Runtime Web 1.30.0 [1], Chrome on macOS (user agent recorded), cross_origin_isolated = false, WASM threads 1. Each requested provider has 5 page/session starts × 100 timed requests after 10 excluded warm-ups (Table 9).

Reading rules fixed by the summaries themselves: requested_is_proven = false—the harness records only the requested provider and session creation, not graph-partition status, so WebGPU execution is unverified and no speed ratio between the columns is claimed; process_cold_claimable = false—“one fresh page/session per source report; browser caches may persist”, so these are not independent browser process launches; JS heap before/after are two observations, not a peak; process RSS and GPU memory are UNAVAILABLE. Page/session load time ranged 1133–1337 ms across the five WASM page starts. These runs used historical full graphs, not the corrected SDK or the shared-v1 bundle, and they are not evidence of current browser latency.

Browser numerical parity: not established. The internal review record (not public) of the browser-parity attempt records status incomplete_report_not_recovered, correctness_pass = null and latency_ms = null: parity attempts stalled, no completed raw report was recovered, and unit or mocked tests do not substitute. The stalled run is not a pass, and no Diffusion browser timing exists.

6 External Evaluation

6.1 Mind2Web: original test splits, exact-k Direct selection (negative result)

Acquisition. osunlp/Mind2Web [6] at dataset revision 17ece8eb89862368edc0cc806acee6fca5163474 (repository revision 33bd95caeee7bba22dd08ecc935845e15c5e5dc7), licence CC-BY-4.0, usage evaluation-only, no_training = true, no_unzipped_redistribution = true; test archive 567,745,122 bytes, SHA-256 8f5f8e72afab942fe97cdf7fb397e179885d89b5c16862288e9a14bc6d41ca89, status downloaded-and-hash-verified (MEASURED). Raw and normalized data, retrieval text and per-row predictions stay local; the packet carries their digests and the aggregate reports.

Table 10. All twelve Mind2Web split/k cells (MEASURED; rounded to four decimals, full precision in `reports/mind2web/evaluation-replay-v1/summary.json`). “Overflow” is a strict 512-token budget overflow recorded as an error; “abstain” is a confidence abstention. Conditional selection accuracy (task-macro) is 0.0 in every cell and complete-case selection accuracy is null (undefined) because no selection was accepted.

Split	k	Rows	Positive	Recalled	Recall micro	Recall task-macro [95% CI]	Cond. acc.	Overflow	Abstain	Accepted
Cross-Task	5	2,094	1,975	96	0.0486	0.0713 [0.0549, 0.0894]	0.0	1,246	848	0
	10	2,094	1,975	173	0.0876	0.1199 [0.0983, 0.1429]	0.0	1,867	227	0
	20	2,094	1,975	331	0.1676	0.2100 [0.1833, 0.2397]	0.0	2,081	13	0
	32	2,094	1,975	497	0.2516	0.2916 [0.2623, 0.3223]	0.0	2,092	2	0
Cross-Website	5	1,373	1,314	80	0.0609	0.0806 [0.0615, 0.1025]	0.0	655	718	0
	10	1,373	1,314	128	0.0974	0.1240 [0.0992, 0.1489]	0.0	1,182	191	0
	20	1,373	1,314	234	0.1781	0.2216 [0.1908, 0.2537]	0.0	1,364	9	0
	32	1,373	1,314	361	0.2747	0.3210 [0.2860, 0.3561]	0.0	1,370	3	0
Cross-Domain	5	5,911	5,590	390	0.0698	0.0916 [0.0819, 0.1027]	0.0	3,180	2,731	0
	10	5,911	5,590	648	0.1159	0.1447 [0.1328, 0.1587]	0.0	4,980	931	0
	20	5,911	5,590	1,071	0.1916	0.2248 [0.2099, 0.2405]	0.0	5,852	59	0
	32	5,911	5,590	1,544	0.2762	0.3021 [0.2869, 0.3188]	0.0	5,900	11	0

Protocol. Scope per the replay summary: “Replay of saved exact-k action-target predictions over the released candidate pool; no model execution, test-set adaptation, or browser task success.” Prediction scope: “exact-k Direct action-target selection; strict 512-token budget; no calibration; no training”. Retrieval uses the released candidate pool (BM25 [7]) with teacher-forced prior action history—not end-to-end DOM retrieval and not browser task success. Selection uses the uncalibrated shared Direct head with a strict 512-token budget. Bootstrap intervals use 1,000 draws over `task_id_or_example_id`. All 12 expected cells are included (`matrix.complete = true`); Table 10 and Figure 2 show every cell.

Totals. 37,512 attempts; 31,769 strict token-budget overflows recorded as errors; 5,743 confidence abstentions; 0 accepted selections (DERIVED sums over the twelve cells). Because no selection was accepted on any recalled row, conditional selection accuracy and acceptance coverage are 0.0 in every cell and complete-case selection accuracy is null (undefined, not zero). Candidate recall is a property of the

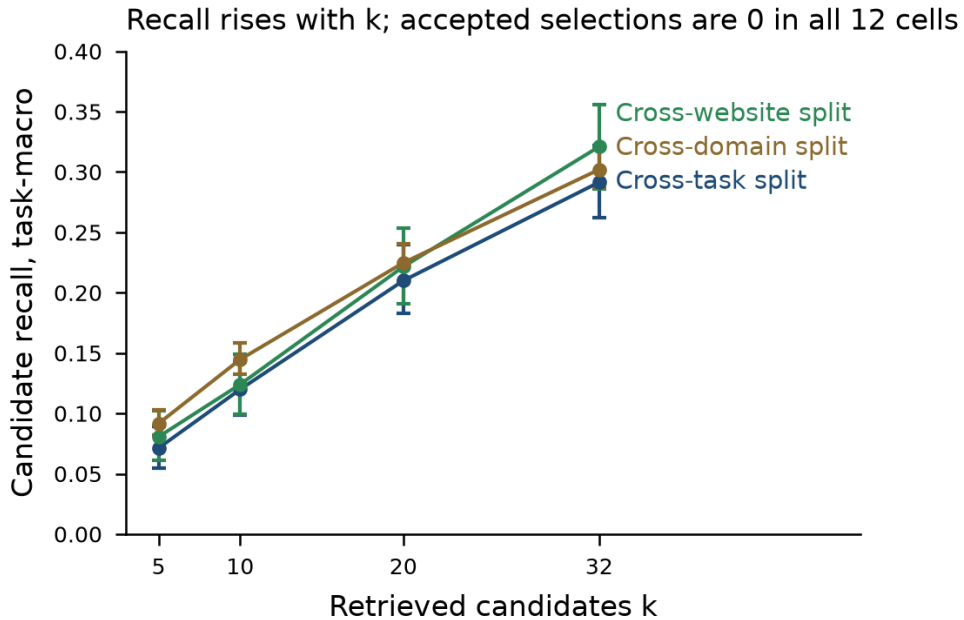


Figure 2. Candidate recall (task-macro, 95% task-bootstrap CI) by k for the three official splits; accepted selections are zero in every cell.

Table 11. Historical local-model screen (MEASURED (HISTORICAL); 200 synthetic rows each; no new model calls). Role names are metadata: the three runs share identical prompt sets and one generic system instruction.

Model (local tag)	Role (metadata)	Rows	Known-label acc.	Distinct prompts	Accepted for training
qwen3.8:27b-mlx	writer	200	0.00	10	0
gemma4:31b	judge	200	0.00	10	0
muse-glimmer:30b-mlx	analyst	200	0.09	10	0

released candidate pool at each k, not of the Vons head. The dominant failure is that Mind2Web states exceed the 512-token budget at larger k; the remaining attempts fall below the confidence threshold. This is a valid bounded negative result for the current contract limits; it is not an end-to-end web-agent evaluation. It remains the historical baseline (84.6902% strict overflow). Post-freeze status: a complete v2 matrix over the same three splits and four values of k, with a label-free structured DOM compression of the input, is reported separately in Section 8.4 (POST-FREEZE (HANDOFF)).

6.2 Jev / TypeSafe four-workflow comparison: not run

Status `not_run` with `agreement = null` (UNAVAILABLE). `docs/JEV_COMPARISON_PROTOCOL.md` freezes the comparison basis (four workflows; Noul/Choice/Score primitives; reference answers averaged from two frontier models; structured wrapper for LLM baselines) and lists the frozen inputs required before any run can be called equivalent. Exact corpus, reference outputs, scorer, provider settings and cost conditions are unavailable; any adaptation to Vons limits must be labelled as adapted. Vendor-published figures [8], [9] are vendor claims and are not reproduced here.

7 Local-Model Suitability Screen and Contributor Attribution

7.1 Screen and alignment audit

All 600 historical responses (200 per model, sequential on one Mac) are in the packet under `artifacts/ollama/`, and `reports/local-model-alignment-v1.json` audits them read-only with `new_model_calls = 0` and `teacher_outputs_used_for_training = false` (MEASURED (HISTORICAL); Table 11).

The audit’s findings: every answerable pilot label is `call_tool` while all three models selected `respond_directly`; the saved prompts do not identify a required tool or a rule distinguishing those two actions, so the low accuracies are a **prompt/label-specification limitation, not a measured model error rate**; Qwen and Gemma chose `clarify` on the 18 underspecified rows where the label requires abstention (a mapping the prompt did not state); Muse’s 0.09 comes from its 18 abstention booleans while keeping a non-null choice, which conflicts with the public abstention contract; role names are metadata, not swapped-role evaluations. No general ranking, role qualification or teacher-data acceptance follows; no local model was admitted as a training teacher.

7.2 Attribution

Models used. Qwen (`qwen3.8:27b-mlx`), Gemma (`gemma4:31b`) and Muse Glimmer (`muse-glimmer:30b-mlx`) were used in writer-, judge- and analyst-role suitability benchmarks; responses are preserved for comparison, not accepted as labels or as calibrated confidence. Muse Glimmer’s official upstream card and LICENSE state Apache-2.0 and the repository also carries a separate usage policy [10]; the local Ollama conversion exposes no licence field and no mapping to an upstream revision, so the **local conversion’s upstream revision remains unverified** and the historical exclusion flag in the raw record is retained as history. Shared encoder/tokenizer: `google/bert_uncased_L-4_H-256_A-4` [11], [12] at revision `387825ce42dbb39b87911cdf8e383ee3b25184f8`.

Contributors, attributed only to recorded outputs. **Claude Code**—numeric fixes to `vons/evaluation.py`, `vons/calibration.py` and their tests (selective-risk denominator, Brier definition, null/finite

validation, calibration eligibility), with scoped checks in an internal review record (not public). **TRAE**—`tools/summarize_browser_benchmark.py` and its tests, and a local licence inventory, accepted with an integrator disposition that grants no redistribution clearance. **Qwen (Ollama)**—one bounded source-only review of the Mind2Web predictor with three findings dispositioned as not reproduced / rejected (existing guard) / accepted diagnostic improvement; the disposition records “One bounded static review, not validated general coding expertise.” **Codex**—integration, shared-bundle export, Mind2Web import/prediction/replay, packet freeze and claim ledger. Writing assistance for this report is disclosed in Section 13.

8 Post-freeze Remediation: v2 Evidence (2026-09-25)

Everything in this section carries the label `POST-FREEZE (HANDOFF)`. It describes post-freeze evidence, 2026-09-25 (Codex-verified handoff; not independently rerun or accessed by Claude Science). The evidence is indexed by `reports/goal-v2/evidence-summary.json` (SHA-256 `39f53771bbf4ffe454dbac09412baf0f371804e6442bb197eaf9879afbfc640a`); the handoff supplied, for each remediation goal, the file paths, their SHA-256 digests and the values reproduced here, and the drafting session copied those values verbatim without opening, hashing, rerunning or otherwise accessing any of the files. **None of these paths is a member of the frozen packet** `publication-evidence-v2.zip` (Section 1; one stated digest coincides with a frozen member, Section 8.1), and nothing in Section 4 to Section 7 is altered by this section: the frozen v1 results remain the historical baseline. Four remediation goals were addressed; Table 12 places each against its frozen baseline, and Section 8.1 to Section 8.4 give the handoff values with their scopes. The same four-goal structure is used in the companion paper.

Table 12. The four remediation goals: frozen v1 baseline (left; packet values) versus post-freeze v2 evidence (right; `POST-FREEZE (HANDOFF)`, handoff values copied verbatim). The two columns rest on different sample sizes, data sets, code states and—for goal 4—different inputs, and are not combined into ratios or ranges. No row implies external generalization, browser performance or deployment readiness.

Goal	Frozen v1 baseline (MEASURED, 2026-09-24)	Post-freeze v2 (POST-FREEZE (HANDOFF), 2026-09-25)
1. Padding and parity	Fixed padding to 32 slots $\times$ 512 tokens; sequence-padding cost $C - B = 204.373$ ms at inference + readback p50 on CPU (240 samples per cell; Table 8); shared-v1 CPU partition parity 80 / 80 at atol = rtol = 10^{-5} (Section 5.2)	Per-request padding to the longest live tokenized candidate, bounded by the manifest budget and 512, in Python and TypeScript; one dynamic graph executed widths 64 and 128 with finite, identical live outputs; fresh local CPU benchmark, 80 samples per cell, zero failures: Direct 32 slots $\times$ 64 tokens inference/readback p50 19.750145 ms (< 20 ms gate passes narrowly); Direct 4 live slots $\times$ 64 tokens p50 2.672417 ms; Direct 32 slots $\times$ 512 tokens p50 263.363521 ms; sequence-padding delta p50 243.613376 ms; frozen shared-v1 re-verified read-only, 20 cases, 80 samples, max abs differences 0.0, pass pilot-v2: 2,000 rows, 154 unanswerable; answerable labels <code>call_tool</code> 462, <code>clarify</code> 462, <code>refuse</code> 461, <code>respond_directly</code> 461; 200-row test split with 184 answerable (46 / 47 / 46 / 45) and 16 unanswerable rows; vector scaling <code>vons.calibration/v3</code> with strictly interior per-class temperatures 0.734359 / 0.741111 / 0.810072 / 0.669444; fitted abstention threshold 0.0 (Brier-based fit, not a hard-coded gate); calibrated Direct test: answerable accuracy 1.0, macro-F1 1.0 (constant-majority macro-F1 0.101732), coverage 0.92, selective risk 0.0, Brier 2.760749e-6, NLL 0.00091409, ECE 0.036557
2. Data and calibration	Single-label split: all 182 answerable rows <code>call_tool</code> (constant predictor reaches accuracy 1.0); temperature 0.05 at its search lower bound; calibrated test coverage 0.91, selective risk 0.0, ECE 0.2233 (Table 6)	Token-level cross-attention over full encoder sequence states, cosine noise schedule, 64 training diffusion steps, 8 fixed inference steps, cross-entropy over continuous label-smoothed soft targets; twelve strictly decreasing epoch mean losses $1.276957 \rightarrow 0.439098$; ONNX export with dynamic option and sequence axes, checker passed, finite ORT execution at four shapes; synthetic test: answerable accuracy and macro-F1 1.0 (random-choice baseline 25%), coverage 0.92, selective risk 0.0,
3. Diffusion head	Pooled-only conditioning, weak terminal-noise schedule, four fixed DDIM steps; answerable accuracy 0.011–0.060 at coverage 0.11–0.29, selective risk 0.810–0.909 (Table 5)	

Goal	Frozen v1 baseline (MEASURED, 2026-09-24)	Post-freeze v2 (POST-FREEZE (HANDOFF), 2026-09-25)
4. Mind2Web input	Full candidate text with teacher-forced history; 37,512 attempts, 31,769 strict 512-token overflow errors (84.6902%), 5,743 confidence abstentions, 0 accepted; conditional selection accuracy 0.0 (Table 10)	Brier 0.0136773, NLL 0.106455; decision-head-only p50 2.559438 ms Label-free structured DOM compression (presentation attributes removed; semantic tag / text / type / aria-label / id kept; ≤ 15 tokens per candidate; last 2 history actions; exact-token hard cap ≤ 512); complete exact-k matrix over 3 splits $\times k \in \{5, 10, 20, 32\}$; 37,512 attempts: 37,512 accepted, 0 errors, 0 abstains, 0 overflow (0.0%); prompts 52–512 tokens (maximum original prompt 185,879 tokens); replay: 5,553 recalled rows, 323 correct, weighted micro conditional selection accuracy 0.0581668 under an explicitly uncalibrated 0.55/k + 0.5 acceptance gate

8.1 Goal 1: Dynamic ONNX padding and re-verified parity

Implementation (handoff-stated). The Python package and the TypeScript SDK now pad each request to the longest live tokenized candidate, bounded by the manifest budget and by 512 tokens; the TypeScript timing record’s `sequence_length` field records the actual feed width. A real ONNX Runtime CPU regression executed one dynamic graph at widths 64 and 128 with finite and identical live outputs.

Fresh local CPU shape benchmark (reports/onnx-shape-decomposition-v2.json, SHA-256 10d1f962d7809b87d4aef64a00ae5804c6cb2aaf1ad037120e1eede1989c1c0a; 80 samples per cell, zero failures). Table 13 lists the handoff values. The < 20 ms acceptance gate for the Direct 32 slots $\times$ 64 tokens cell (DESIGN TARGET) passes narrowly at p50 19.750145 ms. These are local CPU ONNX Runtime reference values on one machine, not a universal deployment guarantee, and not browser values. They are **not** placed in the same table as the frozen four cells (Table 8): the frozen cells used 240 samples per cell (3 session starts $\times$ 20 repeats $\times$ 4 rows) under the fixed-padding graphs, whereas the v2 cells used 80 samples per cell under the dynamic-padding code; no $v1 \rightarrow v2$ speed ratio is computed in this report.

Read-only re-verification of the frozen shared-v1 partition (reports/shared-v1/parity-v2-readonly.json, SHA-256 d680d0f7f1b02f7dc10b39b40b6ff4c6341fe7cc1c317cb4f0a256cbe63b6d5d). The frozen shared-v1 bundle was re-verified, not modified: 20 cases, 80 samples, maximum absolute score and answerability differences 0.0, pass. The scope remains graph-partition parity only—no quality, browser, calibration or latency claim follows, exactly as for the frozen parity file (Section 5.2). **Digest coincidence (checked against the frozen packet).** The handoff-stated digest of parity-v2-readonly.json is identical to the SHA-256 of the frozen packet member reports/shared-v1/parity.json (MEASURED: recomputed from the packet in the drafting session). The frozen file carries no timestamp or run identifier (schema vons.shared-bundle-parity/v1; fields: manifest digest, provider, ORT version, threads, tolerances, 20 cases, 80 samples, pass), so a deterministic re-run that reproduces every value would yield a byte-identical file; the coincidence is therefore consistent both with such a re-run and with a copy of the frozen report, and the drafting session cannot distinguish the two. The re-verification claim rests on the handoff statement, not on the digest.

Table 13. Post-freeze local CPU shape benchmark (POST-FREEZE (HANDOFF)); handoff values verbatim, full handoff precision; zero failures). Local CPU ONNX Runtime reference only; separate from the frozen 240-sample cells of Table 8.

v2 cell (Direct head)	Infer.+readback p50 (ms)	Samples
32 slots $\times$ 64 tokens	19.750145	80
4 live slots $\times$ 64 tokens	2.672417	80
32 slots $\times$ 512 tokens	263.363521	80
Sequence-padding delta (32 $\times$ 512 minus 32 $\times$ 64)	243.613376	80 per cell

8.2 Goal 2: Balanced pilot-v2 synthetic data and vector-scaling calibration v3

Data. pilot-v2 has 2,000 rows, of which 154 are unanswerable. The answerable label counts are `call_tool` 462, `clarify` 462, `refuse` 461 and `respond_directly` 461. Its 200-row test split has 184 answerable rows with counts 46, 47, 46 and 45 respectively and 16 unanswerable rows. Unlike the frozen single-label split (Section 4.1), this split can distinguish a learned predictor from a constant one: the constant-majority macro-F1 on this test distribution is 0.101732 (handoff-stated).

Calibration v3. Vector scaling `vons.calibration/v3` fits per-class temperatures and biases with bounds $[0.1, 10]$ and a log-temperature L2 penalty of 0.01. The fitted temperatures are `call_tool` 0.734359, `clarify` 0.741111, `refuse` 0.810072 and `respond_directly` 0.669444, all strictly interior to the bounds—in contrast to the frozen v2 fit, whose single temperature stopped at its lower bound (Table 6). Dynamic abstention is fitted from the calibrated multiclass Brier score: target Brier 2.294946e-6, resulting confidence threshold 0.0, answerability threshold 0.5, over 185 covered fit rows of 200. **The threshold 0.0 is a fitted result of the Brier-based dynamic-abstention procedure on this data, not a hard-coded confidence gate;** it must not be read as “no abstention mechanism”, and it is specific to this calibration split.

Calibrated Direct test report (SHA-256 156c0ce009271f93f09f6e778e28248e97fd7c8edd1389886d04ab75990189d7). Table 14 gives the handoff values. These are controlled, deterministic, synthetic rule-following results; they are not external generalization, and they are not comparable row-for-row with Table 4 or Table 6, which rest on the frozen single-label split and on a different calibration method.

8.3 Goal 3: Token-conditioned Diffusion redesign

Design (handoff-stated). The new head uses token-level cross-attention over the full encoder sequence states (the frozen head conditioned on the pooled vector only), a cosine noise schedule, 64 training diffusion steps, 8 fixed inference steps, and cross-entropy over continuous label-smoothed soft targets. Twelve epoch mean losses are strictly decreasing: 1.276957, 0.974283, 0.612476, 0.469388, 0.451371, 0.447468, 0.444845, 0.443083, 0.441710, 0.440707, 0.439755, 0.439098.

Export. Export manifest SHA-256 bf8a7480c43c0895901a2f30c83316a3cbcdc6aaa023e3d2a8f363623a5b5aab; ONNX graph SHA-256 7de207c298766ee8eca6921e5f659e33b89689cb6dce38df0f7342bb6339d5c4. The ONNX checker passed; the graph has dynamic option and sequence axes; ONNX Runtime execution was finite at shapes $[1, 4, 64]$, $[1, 4, 128]$, $[1, 8, 64]$ and $[1, 2, 512]$. No shared-bundle partition, bundle-size measurement, CPU parity report or browser measurement for this head is stated in the handoff (UNAVAILABLE for this version).

Synthetic test report (SHA-256 b9db4e59e7300c87dea55b17976f3fbc6b15e9dd25bda9d69f35bb6b490517c1). Answerable accuracy and macro-F1 1.0 versus a 25% random-choice baseline; coverage 0.92; selective risk 0.0; Brier 0.0136773; NLL 0.106455; decision-head-only p50 2.559438 ms. The latency figure is

Table 14. Calibrated Direct head on the pilot-v2 synthetic test split (POST-FREEZE (HANDOFF); handoff values verbatim). Synthetic rule-following data only; not external generalization.

Quantity (pilot-v2 test split, 200 rows)	Value	Label
Answerable accuracy	1.0	POST-FREEZE (HANDOFF)
Macro-F1	1.0	POST-FREEZE (HANDOFF)
Constant-majority macro-F1 on this test distribution	0.101732	POST-FREEZE (HANDOFF)
Coverage	0.92	POST-FREEZE (HANDOFF)
Selective risk	0.0	POST-FREEZE (HANDOFF)
Brier (multiclass, summed over options)	2.760749e-6	POST-FREEZE (HANDOFF)
NLL	0.00091409	POST-FREEZE (HANDOFF)
ECE	0.036557	POST-FREEZE (HANDOFF)

a decision-head-only quantity in the sense of Section 4.5 (encoder excluded, device and provider as recorded by the harness) and is neither a request latency nor a bundle or browser latency; it is not compared with the frozen head-only field (0.267 ms) because the heads, step counts and code states differ. **This is synthetic-only evidence:** it shows that the redesigned head learns the pilot-v2 rule where the frozen four-step head did not learn the frozen rule; it is not ranked against Direct, and no general Diffusion superiority or inferiority follows.

8.4 Goal 4: Mind2Web context compression

Historical baseline (unchanged). 37,512 attempts, 31,769 strict 512-token overflow errors (84.6902%), 5,743 confidence abstentions, 0 accepted (Table 10, MEASURED).

Compression (handoff-stated). A label-free structured DOM compression removes presentation attributes; keeps the semantic tag, text, type, aria-label and id; caps each candidate at 15 tokens; keeps only the last 2 history actions; and applies an exact-token hard cap ≤ 512 . The v2 exact-k matrix is complete across the 3 official held-out splits $\times k \in \{5, 10, 20, 32\}$.

Execution outcomes. Across all 37,512 v2 attempts: accepted 37,512, errors 0, abstains 0, overflow 0 (0.0%). Observed prompts: minimum 52 and maximum 512 tokens; maximum candidate 15 tokens; maximum history 2 actions; maximum original (uncompressed) prompt 185,879 tokens. Prediction index SHA-256 55f471d1998ecbf14b8de88a9f22d69b5e47e362dd0c09517f9c91c65ebb04e9.

Replay. The replay (summary SHA-256 95c4464ca5095149c6a6d4805b799fcaa511fef9d3d77f89d82e715c63059fb6) evaluated 5,553 recalled rows with 323 correct selections, a weighted micro conditional selection accuracy of 0.0581668 (greater than the frozen 0.0). Example endpoints: test_task at $k = 5$, 0.21875; test_domain at $k = 32$, 0.0272021. Per-cell values beyond these two endpoints were not supplied in the handoff and are UNAVAILABLE in this version.

Two scope statements. First, the v2 acceptance uses a candidate-normalized 0.55/k confidence threshold plus the 0.5 answerability threshold and is **explicitly uncalibrated**; zero abstentions is therefore an **execution-enablement result**—every attempt now fits the budget, reaches inference and yields a decision—and **not a calibrated-risk claim**. Second, the conditional selection accuracy 0.0581668 is **released-candidate action-target selection**, conditional on the retriever having recalled the target; it is not browser task success, the recall ceiling of the released candidate pool (Table 10) is unchanged, and the evaluation-only licence scope of Section 6.1 (no training, no redistribution) applies to the v2 run as stated in the handoff.

8.5 Implementer checks stated in the handoff

The handoff records, as implementer checks that the drafting session did not run and cannot confirm: full Python suite 291 passed plus 50 subtests; Ruff clean on all changed Python files; TypeScript 16 tests passed; benchmark bundle build passed; git diff check passed. No commit or push had been made at the time of the handoff, so no repository revision is bound to any post-freeze value in this version (POST-FREEZE (HANDOFF)). These are the implementer’s checks, not checks performed for this report, and—as for the frozen snapshot (Section 11)—unit tests do not establish browser numerical parity.

9 Provenance and Reproduction

9.1 Input verification record

Table 15 records the checks performed on the packet before drafting; Table 16 lists the roles the manifest assigns to the 131 listed files. The full verification record (archive, packet, spec, per-file and per-claim results) is a private file supplied with the report and is not part of the public source.

Table 15. Input verification of the frozen evidence (performed in the drafting session; verification is not re-execution).

Item	Result
Archive publication-evidence-v2.zip	SHA-256 f2fb06c2275a60eec6c5b89938123da599b9979bb10d7e65a5667f6ca69152c4—matches the requested digest
packet.json	SHA-256 7cc5b19cd0e7691919d632d431ff1101d5a5b26a7c11c104c9fa83c132cd5318—matches the requested digest; schema vons.publication-evidence/v2
input-spec.json	SHA-256 matches packet.spec_sha256 (0e464db6cf0c935dbe5818753117c13b01dadbdabcc32de2eef5841eeb4c31)
Listed files	131 / 131 present with matching SHA-256 and byte count; 0 mismatches. The ZIP has 133 members: the 131 listed files plus packet.json (self; verified by its expected digest) and input-spec.json (covered by spec_sha256)
Claim ledger reports/publication-claims-v2.csv	258 / 258 rows re-resolved (JSON pointer, operation, value, source SHA-256) with 0 failures
Code revision	git_revision = null, git_revision_reason “workspace has no Git repository”—frozen capture state at the 2026-09-24 freeze; no later repository commit is bound to any measurement
Independent re-execution	None; restricted Mind2Web data and model weights are not in the packet

Table 16. Roles assigned by packet.json to the 131 listed files.

Role	Files
source_or_config_snapshot	65
measured_report_or_linked_evidence	14
external_evaluation_aggregate	14
source_or_documentation_snapshot	12
model_or_runtime_provenance_weights_not_included	7
review_or_execution_record_not_independent_truth	7
historical_predictions_metrics_superseded_by_v2	6
historical_local_model_screen_not_accepted_training	3
synthetic_evaluation_input	3

9.2 Post-freeze handoff record (not in the packet; not accessed)

Table 17 lists the repository paths and SHA-256 digests stated in the 2026-09-25 handoff for the post-freeze evidence, copied programmatically from the handoff text (never retyped). Every row is `POST-FREEZE (HANDOFF)`: the digest is the integrator’s statement, the drafting session did not open or hash the file, and no independent re-execution took place. The claim ledger records each post-freeze handoff value with the verification note “HANDOFF: value and hash as supplied by Codex on 2026-09-25; file not accessed” (Appendix A); the digest-equality row PF-017 instead carries a split-provenance note stating that `parity-v2-readonly.json` was not accessed and that the frozen member `parity.json` was accessed from the packet and its digest independently recomputed. The frozen packet, its 133 members and Table 15 are unchanged.

9.3 Reproduction commands (audited against the frozen argparse definitions; not executed)

Every command below was checked against the `add_argument` definitions in the snapshot’s `tools/*.py` and `vons/cli.py`. The post-freeze code state (dynamic padding, `vons.calibration/v3`, the token-conditioned Diffusion head, the Mind2Web compression) is **not** in the snapshot; no command for producing the post-freeze evidence was audited, and none is listed here (`UNAVAILABLE` for this version). Angle-bracket tokens are **placeholders** (`<PATH>`: a filesystem path, `<NAME>` a string, `<HEX>` a digest); square brackets mark optional arguments; slash-separated lists are the parser’s allowed choices. None of these commands was run for any version of this report.

C1. Recompute `vons.metrics/v2` from saved prediction reports (`--report` is repeatable; source of `quality-summary-v2.json`).

Table 17. Post-freeze evidence files as stated in the Codex-verified handoff (POST-FREEZE (HANDOFF)). Not members of publication-evidence-v2.zip; not accessed or hashed in the drafting session.

Handoff file	Stated SHA-256 (POST-FREEZE (HANDOFF))
reports/goal-v2/evidence-summary.json (primary index)	39f53771bbf4ffe454dbac09412baf0f 371804e6442bb197eaf9879afbfc640a
reports/onnx-shape-decomposition-v2.json	10d1f962d7809b87d4aef64a00ae5804 c6cb2aaf1ad037120e1eede1989c1c0a
reports/shared-v1/parity-v2-readonly.json	d680d0f7f1b02f7dc10b39b40b6ff4c6 341fe7cc1c317cb4f0a256cbe63b6d5d (identical to the digest of the frozen member reports/shared-v1/parity.json; Section 8.1)
Calibrated Direct test report (pilot-v2; path not stated in the handoff)	156c0ce009271f93f09f6e778e28248e 97fd7c8edd1389886d04ab75990189d7
Diffusion export manifest (path not stated)	bf8a7480c43c0895901a2f30c83316a3 cbcdc6aaa023e3d2a8f363623a5b5aab
Diffusion ONNX graph (path not stated)	7de207c298766ee8eca6921e5f659e33 b89689cb6dce38df0f7342bb6339d5c4
Diffusion synthetic test report (path not stated)	b9db4e59e7300c87dea55b17976f3fbc 6b15e9dd25bda9d69f35bb6b490517c1
Mind2Web v2 prediction index (path not stated)	55f471d1998ecbf14b8de88a9f22d69b 5e47e362dd0c09517f9c91c65ebb04e9
Mind2Web v2 replay summary (path not stated)	95c4464ca5095149c6a6d4805b799fca a511fef9d3d77f89d82e715c63059fb6

```
python tools/summarize_quality.py --data <PATH: test jsonl> --report <PATH: prediction report> [--report <PATH> ...]
--output <PATH: summary json> --version v2
```

C2. Temperature-then-threshold calibration on the disjoint calibration split. Recorded paths: data/generated/pilot-calibration.jsonl, reports/pilot-direct-calibration.json, data/generated/pilot-test.jsonl, reports/pilot-direct-test.json → reports/pilot-direct-test-calibrated-v2.json.

```
python -m vons.cli calibrate --calibration-input <PATH: calibration jsonl> --calibration-report <PATH: calibration
report> --test-input <PATH: test jsonl> --test-report <PATH: test report> --output <PATH: calibrated report> [--
target-risk 0.05]
```

C3. Partition the two full graphs into one encoder and two heads; writes the shared manifest and, if requested, the CPU parity report.

```
python tools/export_shared_bundle.py --direct <PATH: Direct full graph .onnx> --diffusion <PATH: Diffusion full
graph .onnx> --tokenizer <PATH: tokenizer dir> --output <PATH: new bundle dir> [--direct-calibration <PATH>] [--
parity-report <PATH>]
```

C4. G0 bundle inventory with SHA-256 per asset and the 64 MiB model-asset check.

```
python -m vons.cli build-bundle-manifest --onnx <PATH> --tokenizer <PATH: dir> --output <PATH: manifest json> [--
checkpoint <PATH>] [--model-manifest <PATH>] [--calibration <PATH> ...] [--runtime-asset <PATH> ...] [--license
<PATH> ...] [--config <PATH> ...] [--source-root <PATH>] [--source-file <PATH> ...]
```

C5. Independent re-enumeration of a bundle manifest.

```
python -m vons.cli verify-bundle-manifest --manifest <PATH: manifest json>
```

C6. Direct full-bundle versus checkpoint CPU parity.

```
python tools/verify_onnx_bundle.py --checkpoint <PATH: model.pt> --bundle <PATH: full .onnx> --output <PATH: parity
json>
```

C7. Four-cell CPU padding decomposition; the numeric values are those recorded in onnx-shape-decomposition.json. The parser defaults differ from the recorded run (session-starts 1, warmup 2, repeats 5).

```
python tools/benchmark_onnx_shapes.py --direct-manifest <PATH> --diffusion-manifest <PATH> --input <PATH: jsonl> --limit 4 --session-starts 3 --warmup 5 --repeats 20 --seed 7 --live-candidates 4 --short-sequence 64 --full-sequence 512 --output <PATH: report json>
```

C8. Evaluation-only import of the pinned official Mind2Web test archive. `--split` takes one or more official split names, or `all` alone—all cannot be combined with another split (the parser errors otherwise); `--extract-archive` takes a path; restricted data are not in the packet.

```
python tools/import_mind2web.py --input-dir <PATH: acquisition dir> --split <one or more of test_task test_website test_domain, or all alone> --output-dir <PATH: restricted output dir> --source-manifest <PATH: acquisition manifest> [--extract-archive <PATH: test.zip>]
```

C9. Exact-k Direct selections without reading labels. With the default `--backend direct` (the backend of every recorded run) **both `--manifest` and `--manifest-sha256` are required** and the parser errors otherwise (“Direct requires a shared manifest and externally pinned `--manifest-sha256`”); they may be omitted only with `--backend lexical_top1`.

```
python tools/predict_mind2web.py --retriever <PATH: retrieval rows> --output <PATH: predictions> --split <test_task / test_website / test_domain> --k <5 / 10 / 20 / 32> [--backend direct] --manifest <PATH: shared manifest> --manifest-sha256 <HEX>
```

C10. Frozen evaluator for one predictions file.

```
python tools/evaluate_mind2web.py --rows <PATH: gold rows> --predictions <PATH> --output <PATH> [--split <NAME>] [--k 5 10 20 32]
```

C11. Hash-checked replay of the 12 cells (source of `evaluation-replay-v1/`).

```
python tools/replay_mind2web_evaluation.py --rows <PATH: gold rows> --run-index <PATH: run index json> --predictions-dir <PATH> --output-dir <PATH: replay output dir> [--allow-subset]
```

C12. Merge independently saved one-session browser reports.

```
python tools/merge_browser_benchmarks.py --report <PATH: session report> [--report <PATH> ...] --output <PATH: merged json>
```

C13. Validate and recompute a merged browser report.

```
python tools/verify_browser_benchmark.py --report <PATH: merged json> --output <PATH> [--allow-incomplete]
```

C14. Per-session counts and p50/p95 summary (source of the two `*-trae-summary.json` files); exactly one of `--input` or `--report-input` is given.

```
python tools/summarize_browser_benchmark.py (exactly one of --input <PATH: samples jsonl> or --report-input <PATH: merged json>) --output <PATH: summary json>
```

C15. Browser-versus-CPU parity verifier (harness exists; no completed browser matrix report exists).

```
python tools/verify_browser_parity.py --report <PATH: browser trace> --manifest <PATH: bundle manifest> --output <PATH> [--atol 1e-05] [--rtol 1e-05]
```

C16. Read-only alignment audit of the saved local-model responses (`--input` is repeatable, one raw record per model).

```
python tools/audit_local_model_alignment.py --input <PATH: raw ollama record> [--input <PATH> ...] --output <PATH: audit json>
```

C17. Freeze the allowlisted files into a new directory, writing `packet.json` inside it and a sibling `<directory>.zip`; refuses to overwrite an existing directory or ZIP.

```
python tools/freeze_publication_packet.py --spec <PATH: input-spec.json> --output <PATH: NEW directory>
```

C18. Derive the field-level claim ledger from the frozen specification.

```
python tools/build_claim_ledger.py --spec <PATH: configs/publication-claims-v2.json> --output <PATH: claims csv>
```

C19. Training, evaluation and export entry points exist with these optional arguments; the exact argument values used for the pilot runs are UNAVAILABLE in the snapshot.

```
python -m vons.cli train / evaluate / export / export-bundle / export-diffusion-bundle [--config <PATH>] [--model <NAME>] [--input <PATH>] [--checkpoint <PATH>] [--output <PATH>]
```

C20. Integrator checks recorded 2026-09-24: 241 tests / 50 subtests, lint exit 0, 10 SDK tests (DOCUMENTED STATUS); not re-executed for this report.

```
.venv/bin/pytest -q; .venv/bin/ruff check .; npm --prefix sdk/typescript test
```

The training/export configurations are configs/pilot.json and configs/pilot-diffusion.json; the argument values used for the recorded pilot runs are UNAVAILABLE in the snapshot. Model weights (model.pt, .onnx.data) are not in the packet; the bundle manifests carry their digests. Historical browser sessions are reproducible only from the merged reports' source-report digests; the raw single-session files are not in the packet.

10 Failure Taxonomy: Current State

The class structure in Table 18 was defined in the earlier private Tech Report sessions and is applied **retrospectively** to the v2 packet; the packet contains no record fixing these classes before its measurements were taken, so the table is a post-hoc classification, not a preregistered one. Boundary and provenance failures are counted with identifiers; behavioural and cost failures may be rates with denominators. A class reports zero only when its instrument ran over a stated denominator. The fourth column, new in v1.1, records what the post-freeze handoff states for each class; it is POST-FREEZE (HANDOFF) throughout, does not replace the frozen column, and a zero in it is the implementer's stated count, not a count verified here.

Table 18. Failure taxonomy applied retrospectively to the v2 packet (post-hoc classification), with a post-freeze status column added in v1.1 (POST-FREEZE (HANDOFF); handoff statements, not independently verified). The frozen column is unchanged from v1.0.1.

Class	Instrument	Frozen state (v2 packet, 2026-09-24)	Post-freeze status (POST-FREEZE (HANDOFF), 2026-09-25)
F1 Decision behaviour—contract-invalid output, wrong choice, over/under-abstention, covered unanswerable rows, export parity mismatch	metrics/v2 counts; CPU parity	Direct: 0 wrong answerable choices, 18 covered unanswerable rows per seed (selective risk 0.09) before calibration; 0 after. Diffusion: majority abstention, selective risk 0.81–0.91. Shared-v1 CPU partition parity: 0 mismatches / 80. Browser parity: unavailable (stalled).	Direct on pilot-v2 (calibrated): accuracy 1.0, coverage 0.92, selective risk 0.0; token-conditioned Diffusion on pilot-v2: accuracy 1.0, coverage 0.92, selective risk 0.0 (synthetic only). Shared-v1 read-only re-verification 0 mismatches, max abs diff 0.0 (80 samples). Browser parity: still unavailable .
F2 Boundary / invariant—model-side consent or execution, model-asserted confirm, unregistered tool executed, fail-open risk, non-deterministic derivation, missing audit record	Adapter unit tests; audit-log replay	Unit-level behaviour tested (241 Python tests, 10 SDK tests, integrator-recorded). Audit-log replay: unavailable —no real-workflow adapter log exists; no class reports zero.	Handoff states 291 Python tests + 50 subtests and 16 TypeScript tests (implementer checks). Boundary rules unchanged. Audit-log replay: still unavailable ; no class reports zero.
F3 Data and provenance—split leakage, digest mismatch, unflagged synthetic rows,	Packet digests; provenance manifests	131/131 file digests verified; 258/258 claims resolved. Open: Muse local-conversion upstream	Nine handoff files with stated SHA-256 digests (Table 17), not independently hashed; no

Class	Instrument	Frozen state (v2 packet, 2026-09-24)	Post-freeze status (POST-FREEZE (HANDOFF), 2026-09-25)
licence/usage-policy unresolved, restricted-data leakage		revision; shared-v1 carries no notice payload; checkpoint digests absent from report configurations. Restricted Mind2Web rows correctly absent from the packet.	repository commit bound (no commit or push at handoff). Frozen packet digests unchanged. Open items unchanged.
F4 Cost measurement— conflated scopes, unverified provider, non-independent sessions, missing warm-up policy, memory nulls, cross- device carry-over	Timing reports' own flags	Historical browser: provider unverified (<code>requested_is_proven = false</code>), page starts not process launches, memory null—all disclosed. CPU decomposition: 5 warm-ups excluded, 3 sessions. Earlier 20-row bundle timings superseded.	v2 CPU cells: 80 samples per cell, zero failures, local CPU ONNX Runtime reference; no browser measurement; no v1 → v2 ratio computed. Decision-head-only p50 for the new Diffusion head is head-only, not request latency.
F5 Harness and analysis— held-out contamination, post- hoc thresholds, best-of-seeds, unavailable rendered as zero	Metric policy; ledger	Calibration split disjoint from test; all three seeds reported; <code>null</code> preserved (complete-case accuracy). No preregistered thresholds exist (open).	Abstention threshold 0.0 is a fitted Brier-based result on the calibration split, not a preregistered or hand-set gate; Mind2Web v2 acceptance gate $0.55/k + 0.5$ is explicitly uncalibrated. No preregistered thresholds exist (open).
F6 Direct/Diffusion asymmetry—unequal tuning, unmatched shapes, different packets	Shape decomposition; training records	Shapes matched at 32×512 on CPU; tuning effort not equalized; Diffusion at one step count. Comparison unadjudicated; no ranking.	New Diffusion head trained and tested on pilot-v2 with 64 / 8 steps; Direct calibrated on pilot-v2; no matched-shape CPU cells or equal-effort tuning record for the two heads on pilot-v2. Comparison still unadjudicated; no ranking.
F7 External-evaluation validity—budget overflow, retrieval ceiling, teacher forcing, task-success conflation	Mind2Web execution outcomes	31,769 overflows / 37,512 attempts; recall bounded by the released pool; teacher-forced history; no task-success claim.	0 overflows / 37,512 attempts after label-free DOM compression (≤ 15 tokens per candidate, 2 history actions, ≤ 512 hard cap); conditional selection accuracy 0.0581668 on 5,553 recalled rows; recall ceiling of the released pool unchanged; still teacher-forced history; still no task-success claim.

11 Limitations and Negative Results

1. All quality numbers are synthetic-pilot (200 rows; all 182 answerable rows share the label `call_tool`, so a constant predictor matches Direct's accuracy) except the Mind2Web negative result; no positive external-task result exists.
2. Mind2Web: 0 accepted selections across 37,512 attempts; the 512-token budget overflows on 31,769; conditional accuracy 0.0; complete-case accuracy undefined. Evaluation-only licence scope; no training or redistribution.
3. Diffusion pilot: pooled-only conditioning, weak terminal-noise schedule and decoder limitations; fixed four steps; no general inferiority or superiority conclusion.
4. Calibration: one seed, one head; temperature at the search lower bound; ECE 0.223 after calibration; no general calibration claim.
5. Browser: historical Direct full graph only; provider unverified; not independent process launches; process/GPU memory null; not the current SDK or shared bundle; no completed parity matrix; Score requests unsupported; Diffusion browser timing absent.
6. The shared bundle's size result applies to model assets only; notices and runtime are excluded; release status is "research candidate".
7. No Git revision at the evidence freeze (`git_revision = null`; the freezing workspace was not a Git checkout); a repository created afterwards cannot be bound retroactively to any measurement;

checkpoint digests absent from report configurations; historical runtime code identity is bound only where reports record it.

8. Jev comparison not run; the Chrome-extension channel is a proposal.
9. Local-model screen: prompts did not establish the `call_tool` label rule; no ranking, role qualification or teacher acceptance; Muse local revision unverified.
10. Public contract: the Python character-estimate validator and the TypeScript tokenizer budget are not yet a fully harmonized public contract.
11. The integrator's recorded checks (241 Python tests with 50 subtests; lint exit 0; 10 SDK tests; SDK build exit 0) were not re-run for this report, and unit or mocked tests do not establish browser numerical parity.

Items 1–11 describe the frozen evidence and are unchanged. The following items concern the post-freeze evidence of Section 8 (POST-FREEZE (HANDOFF)) and are new in v1.1.

12. Every post-freeze value rests on the 2026-09-25 Codex-verified handoff: the drafting session did not access, hash, rerun or otherwise verify any of the nine files in Table 17, none of which is in the frozen packet; no repository commit is bound to them (no commit or push at handoff). The handoff's implementer checks (291 Python tests + 50 subtests, Ruff, 16 TypeScript tests, bundle build, git diff check) are the implementer's, not this report's.
13. Post-freeze quality results (calibrated Direct; token-conditioned Diffusion) are on the deterministic, rule-following pilot-v2 synthetic set only; they show that the balanced split can distinguish learning from a constant predictor (macro-F1 1.0 vs 0.101732), not external generalization. The two heads are not ranked against each other; no equal-effort tuning record or matched-shape CPU cell exists for them on pilot-v2.
14. The fitted abstention threshold 0.0 is a result of the Brier-based dynamic-abstention fit on one calibration split (target Brier 2.294946e-6; 185 covered fit rows of 200); it is not a hard-coded gate and is not a general calibration claim.
15. Post-freeze latency values are local CPU ONNX Runtime reference values (80 samples per cell) or decision-head-only harness values (2.559438 ms); the < 20 ms gate passes narrowly on one machine; no browser measurement, provider verification, cold start or memory figure was added, and no v1 → v2 speed ratio is computed because sample sizes, session settings and code states differ.
16. The shared-v1 re-verification is graph-partition parity only, on the frozen bundle; the new Diffusion head has no stated bundle partition, size measurement, CPU parity report or browser measurement.
17. Mind2Web v2: the $0.55/k + 0.5$ acceptance gate is explicitly uncalibrated, so 0 abstentions is an execution-enablement result, not a calibrated-risk result; the conditional selection accuracy 0.0581668 is released-candidate action-target selection conditional on recall, not browser task success; per-cell values other than the two stated endpoints were not supplied; the recall ceiling and the teacher-forced history are unchanged; the historical 84.6902% overflow result remains the baseline.
18. No release gate was re-audited for this version; Table 3 and the hold assessment of Section 12 are the frozen record.

12 Release Assessment

Decision at the evidence freeze: **hold** for public release of the software and models as a research/release assessment; the report itself is suitable for publication as a pilot / negative-result engineering record. The open gates are listed exactly as recorded; version 1.1 of this report closes none of them

and re-audits none of them. The post-freeze handoff (Section 8) reports work that bears on gates G3 and G4 of Table 3 and on none of the browser items below; whether it changes any recorded status is for a future gate re-audit with access to the files, and this version does not announce publication readiness. The author’s separate post-freeze distribution decision is recorded in Section 13; it does not close any gate.

1. Browser numerical parity matrix for Direct and Diffusion on WASM and WebGPU with saved raw reports.
2. Shared-v1 browser loader and calibration parity.
3. Diffusion browser cells, 2/8-step paths, additional seeds, lifecycle/offline and matched padding cells in the browser.
4. Provider-execution evidence and process-level cold start.
5. Memory with explicit scopes or nulls.
6. KASI audit-log replay on real workflow traces.
7. Jev exact inputs or an explicitly adapted benchmark.
8. Preregistered thresholds and analysis plan.
9. Notices inventory for shared-v1 and runtime.
10. Muse local-conversion upstream revision.
11. Licence implementation: at the evidence freeze the free-noncommercial / paid-commercial model was a **proposal only**, the Python package metadata declared Apache-2.0, no new terms were adopted, no open-source claim was made, and upstream conditions were unchanged. The author’s subsequent selection of source-available terms (Section 13) is an `AUTHOR DECISION` whose licence text and notices are not part of the frozen evidence.
12. Cross-review reconciliation of this report against the companion paper (v1.0 consistency check; performed as a separate record).

13 Availability, Licensing and Disclosure

Report. This report and its original figures are licensed under CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>), which includes commercial reuse under its terms. The licence does not cover the Vons software, quoted source excerpts or third-party assets, raw benchmark data, or private verification records (`AUTHOR DECISION`; at the v0.2 stage the article terms were still under owner review).

Software and models at the freeze. At the evidence freeze, the Python package metadata declared Apache-2.0 and a proposed alternative licence had not been adopted; the TypeScript package had no licence field, and the shared-v1 bundle carried no licence or notice payload (`MEASURED: pyproject.toml` and the shared-v1 manifest). The research/release assessment of Section 12 (hold; all scientific gates open) is unchanged.

Distribution decision after the freeze (`AUTHOR DECISION`). After the freeze the author decided on **source-only distribution** of the Vons software under custom source-available terms: noncommercial use is free; enterprise and commercial use require a separate agreement; the terms are not an OSI open-source licence; grants already made under the earlier Apache-2.0 terms are not cancelled. Manuscript links are held back pending review. Source publication does not validate model deployment: none of the open gates in Section 12 is closed by publishing source, and no model weights, browser parity or product readiness is implied. The owner personally performs every commit, push and upload; no assistant performs them. Licence text and notices for these terms are not part of the frozen evidence.

Designated destinations. The author has designated <https://github.com/inlevel9-com/Vons>, <https://huggingface.co/INLEVEL9> and <https://huggingface.co/INLEVEL9/Vons> as destinations for

the source and for any future model card. Their public-access and upload status was **not verified** from the drafting environment, and this report makes no claim about files present at any of them (PLANNED).

Publication and listing status (AUTHOR DECISION status as stated by the author, not re-verified from the typesetting environment). arXiv submission 8127259 is on hold, and this artifact pass does not itself update or publish it. The Chrome Web Store listing for “Vons — Local Decisions” version 0.1.0 is publicly available and was independently verified on 2026-09-25 at <https://chromewebstore.google.com/detail/vons-%E2%80%94-local-decisions/cbmjkkokoojinfgefahncmjfmicinide>. This establishes listing availability only; it is not a scientific result, says nothing about model quality or production readiness, and closes no gate of Table 3 or Section 12.

Data and records. Mind2Web data were used evaluation-only, were excluded from training and are not redistributed; only aggregate reports and hashes are held in the evidence packet. The release status of the synthetic pilot data is undecided. The frozen evidence packet, the claim ledger and the input-verification record are held by the author; the ledger is available for review on request.

Writing assistance. This Technical Report was drafted by Claude Fable 5.1 in the Tech Report sessions from the verified v2 evidence packet, with cross-review by Claude Opus 5.5 (Section 14, correction pass 2) and a bounded post-review correction pass (v1.0.1) requested by the integrator (Codex). Version 1.1 was prepared by Claude Fable 5.1 from the v1.0.1 source and the integrator’s 2026-09-25 handoff of post-freeze evidence; for the post-freeze section the assistant copied the handoff’s paths, digests and values verbatim and did not access, hash or rerun any of the underlying files. The assistants verified digests of the frozen packet, resolved ledger rows, generated the two figures from packet fields and drafted text; they did not generate or alter any measured value, did not train, run inference, benchmark or execute browser tests, and did not modify the paper or any prior artifact. Local models named in Section 7 were subjects of the suitability screen and one bounded code review; none contributed labels or text to this report. The author is responsible for the content, citations and release decisions.

14 Change Log

Each entry states the exact prior location and state of what changed. No measured value, source, model or publication status changed in any correction pass; superseded values appear only here, never in the live results tables.

v0.1 → v0.2 (evidence rebinding, 2026-09-24). Tech Report v0.1 was bound to the earlier research packet (artifacts/research_packet.json, evidence revisions P-1/P-2) and used vons.metrics/v1. Version 0.2 rebinds to publication-evidence-v2.zip and adds the categories measured for the first time in that packet (shared-v1 bundle and CPU partition parity, CPU padding decomposition, historical browser summaries, Mind2Web, local-model alignment audit). Table 19 lists the corrections with their prior state.

Table 19. Change log: v0.1 → v0.2, the two v0.2 correction passes, v0.2 → v1.0, the v1.0.1 correction pass, and the v1.1 additions.

Version	Prior location and state	Change
v0.2	v0.1 §4 results tables: selective risk, Brier and ECE under vons.metrics/v1 (selective risk ignored covered unanswerable rows; Brier divided by option count)	Recomputed under vons.metrics/v2 from the same saved predictions; superseded values moved to Table 20
v0.2	v0.1 CPU bundle latency section: 20-row Direct and Diffusion bundle timings from reports/onnx-direct-latency.json and reports/onnx-diffusion-latency.json	Superseded by the four-cell CPU padding decomposition (Section 5.3); the two latency files are not in the v2 packet and their values are not carried
v0.2	v0.1 calibration: threshold fitted before temperature scaling	Replaced by vons.calibration/v2 (temperature first, then threshold; Table 6); the earlier threshold value is superseded and is not carried

Version	Prior location and state	Change
v0.2	v0.1 §2: risk metadata and confirmed tools described as fields of the decision request	Restated as KASI-side host inputs separate from the general contract (Section 2)
v0.2	v0.1 §2: expected-ordinal Score output described as contract behaviour	Restated as planned evaluation-adaptor behaviour; current SDK returns a discrete choice
v0.2	v0.1 title page: article terms and licence discussed together	Article terms separated from the software licence proposal; both left for owner review at v0.2
v0.2 pass 1	§4.3 sentence “Brier near its maximum”	Removed; measured 0.932–0.937 retained with the v2 range 0–2 stated
v0.2 pass 1	§8.2 reproduction commands as free text	Every command re-audited against frozen <code>add_argument</code> definitions; <code>python</code> prefix on all scripts; explicit placeholders; repeatable flags and choice lists made explicit; <code>freeze --output</code> documented as a new directory
v0.2 pass 1	§9 sentence implying the taxonomy predated the packet	Replaced by an explicit retrospective/post-hoc statement
v0.2 pass 1	Title page and §11: stale “additional authorship” wording	Removed; article terms remained pending
v0.2 pass 2 (R-01)	Abstract, §4.1, §4.2, §10: Direct accuracy 1.0 reported without the constant-label baseline	Stated that all 182 answerable rows are <code>call_tool</code> (read from <code>data/generated/pilot-test.jsonl</code>) so a constant predictor also reaches 1.0; two derived ledger rows added
v0.2 pass 2 (R-02)	§8.2 <code>predict_mind2web.py</code> command showed <code>--manifest/--manifest-sha256</code> as optional	Both shown as required with the default <code>--backend direct</code> ; parser guard cited
v0.2 pass 2 (R-03)	§5.3 derived comparisons (12.289 / 204.373 / 1.289 ms) presented next to the total-request column	Labelled as inference + readback p50 differences; the four cell p50 values cited; four ledger rows added
v0.2 pass 2 (R-04)	§4.2/§4.3 citation ranges ended before the Brier and ECE rows	Ranges extended to include them
v0.2 pass 2 (R-05)	§3.1 test inventory count	Corrected to 19 Python and 3 TypeScript test files, counted from ZIP members
v0.2 pass 2 (R-06)	§4.1 2,000-row source set stated without a source	Cited to <code>artifacts/pilot/model-manifest.json (/data/rows, /data/splits, /data/sha256)</code>
v0.2 pass 2 (R-07)	§1 and §8.1 archive summary	Corrected to 133 members: 131 listed files plus <code>packet.json</code> and <code>input-spec.json</code>
v0.2 pass 2 (R-08)	§8.2 <code>import_mind2web.py --split</code> shown with a single split	Documented as one or more official splits, or <code>all</code> alone; parser guard cited
v0.2 pass 2 (R-09)	§4 ECE-direction caution	Optional finding not applied; caution retained
v0.2 pass 2 (R-10)	§6.1 recalled column without thousands separators	Separators added; PDF title metadata marked “correction pass 2”
v1.0	v0.2 Markdown manuscript with in-text ledger identifiers and an internal review-record path in §5.4 and §7.2	Rewritten in the house format shared with the companion paper (Typst template <code>vons-template.typ</code>); ledger identifiers moved to Appendix A; the path shown as “internal review record (not public)”
v1.0	v0.2 title page: “Article terms remain for owner review”; §11 item 12	Report text and original figures licensed CC BY 4.0 (AUTHOR DECISION); gate wording updated in Section 12 and Section 13
v1.0	v0.2 §11 item 11: licence “proposal only”	Frozen-evidence state retained verbatim; the author’s post-freeze source-available selection added as an AUTHOR DECISION with the paper’s wording
v1.0	v0.2 title page: “Drafting model: Claude Fable 5.1 ... writing assistance only”; §12 disclosure	Consolidated into the writing-assistance paragraph of Section 13, naming the Opus 5.5 cross-review
v1.0	v0.2 §5.3 and §5.4 tables: milliseconds shown to two decimals (e.g. 2.59, 2207.40)	Shown to three decimals (2.586, 2207.400) with a rounding note; source fields unchanged
v1.0	v0.2 §4.4 and §5.4 tables	Added rows already present in the packet and cited by the companion paper: calibration NLL before/after and answerability threshold; browser tokenize p50
v1.0	v0.2 §10 with ten items	Item 11 (integrator checks not re-run) added from §3.1 for agreement with the paper’s limitations
v1.0	v0.2 had no change-log section	This section and Table 20 added; the v0.1 → v0.2 entries are reconstructed from the v0.2 verification and reconciliation records
v1.0	v0.2 §2 prose only	Table 1 added, mirroring the paper’s authority table; no new property introduced
v1.0.1 (F3)	v1.0 §4.1 “Metric definitions”: NLL and Brier named without their population	Defined as means over answerable rows whose prediction has no error and a valid probability vector

Version	Prior location and state	Change
		(abstained predictions with a valid vector included; differs from ECE's covered population), with the per-row formulas; bound to <code>vons/evaluation.py</code> lines 39–46, 186–198 and 241–255 (source-code ledger rows added). Values unchanged
v1.0.1 (F4)	v0.2 §2 and v1.0 §2: “Score questions currently return a discrete choice” (the historical v0.1 → v0.2 entry above is kept as history)	Corrected: the contract schema admits Score questions with a 2–10 item rubric (<code>vons/contract.py</code>); the candidate-selection ONNX head rejects them at runtime (<code>sdk/typescript/src/onnx.ts</code> lines 504–506) and no Python inference backend implements Score inference; expected-ordinal / P(true) mappings remain planned
v1.0.1 (F5)	v1.0 §1 “Code revision”, Table 12 row “Code revision”, Limitation 7: “workspace has no Git repository” stated without time	Qualified as the frozen capture state at the 2026-09-24 freeze (<code>packet.json</code> <code>git_revision = null</code> , <code>git_revision_reason</code>); a repository created afterwards cannot be bound retroactively to any measurement
v1.0.1 (F6)	v1.0 §11 “Decision: hold” and §12 “Software and models” paragraph	Hold kept as the research/release assessment at the freeze; a separately labelled post-freeze author decision added: source-only distribution under custom source-available terms, manuscript links held back, source publication does not validate deployment, owner performs all commits/pushes/uploads; designated destinations named with access status not verified
v1.0.1	Title, date and running header “v1.0”	“v1.0.1” (patch: corrections to existing text; no new measured category, value, source, model, licence or publication status change)
v1.1	v1.0.1 had no post-freeze content; results sections §4–§7 ended with the frozen packet	New Section 8 (POST-FREEZE (HANDOFF)): four-goal table Table 12 and one subsection per goal with the 2026-09-25 handoff values, scopes and caveats; Table 13, Table 14; implementer checks stated as handoff statements. Frozen results in Section 4 to Section 7 unchanged; one “Post-freeze status” cross-reference sentence appended to Section 4.1, Section 4.2, Section 4.3, Section 4.4, Section 5.2, Section 5.3 and Section 6.1
v1.1	§1 label list ended with AUTHOR DECISION	POST-FREEZE (HANDOFF) label defined and added; new paragraph “Post-freeze evidence” stating that the handoff files are not in the frozen packet and were not accessed; “does not claim” list extended
v1.1	Abstract and licence note described the frozen packet only	Post-freeze sentences appended to the abstract, kept separate from the frozen findings; licence note restated for a version that closes and re-audits no gate
v1.1	Table 2, Table 3, Section 3.3 described the frozen snapshot only	Handoff implementer-check row added to Table 2; paragraph after Table 3 stating that the table was not re-audited; sentence in Section 3.3 on handoff-stated dynamic padding
v1.1	§8 provenance covered the packet only	Section 9.2 and Table 17 added (nine handoff digests, copied programmatically); reproduction-command preamble states that no post-freeze command was audited
v1.1	Table 18 had three columns	Fourth column “Post-freeze status (POST-FREEZE (HANDOFF))” added; frozen column unchanged
v1.1	Section 11 had 11 items; Section 12 “closes none”	Items 12–18 added for the post-freeze evidence; release paragraph states that v1.1 closes and re-audits no gate and announces no publication readiness; hold unchanged
v1.1	Section 13 writing-assistance paragraph ended at v1.0.1	v1.1 drafting sentence added: values copied verbatim from the handoff; files not accessed
v1.1	Ledger <code>vons_tech_report_v1_0_claims.csv</code> , 492 rows	<code>vons_tech_report_v1_1_claims.csv</code> : the 492 rows unchanged plus PF-* rows for every post-freeze value (label “POST-FREEZE MEASURED (Codex-verified handoff; not independently rerun by Claude Science)”); Table 21 extended

Version	Prior location and state	Change
v1.1	Title, date and running header “v1.0.1”	“v1.1”; publication readiness not announced; release typesetting pass removed the draft marker from title, header, date and status wording, renamed the ledger to <code>vons_tech_report_v1_1_claims.csv</code> , and added the publication and listing status note in Section 13; no scientific content, value, digest or ledger row changed

Superseded metric values. An earlier metrics version (v1) scored the same saved predictions differently: it ignored covered unanswerable rows in selective risk and divided the Brier score by the option count. Table 20 lists the superseded values next to the v2 values (MEASURED (HISTORICAL)); the v1 values are read from the per-seed report files `reports/pilot-*-test.json`, which the packet lists with role `historical_predictions_metrics_superseded_by_v2`.

Table 20. Superseded v1 and current v2 metric values on the same saved predictions (MEASURED (HISTORICAL) → MEASURED).

Report	Selective risk v1 → v2	Brier v1 → v2	ECE v1 → v2
Direct, seed 7	0.0000 → 0.0900	0.0035 → 0.0139	0.3008 → 0.2163
Direct, seed 17	0.0000 → 0.0900	0.0024 → 0.0097	0.1991 → 0.1154
Direct, seed 27	0.0000 → 0.0900	0.0032 → 0.0129	0.2479 → 0.2881
Diffusion, seed 7	0.8214 → 0.8387	0.2330 → 0.9318	0.4327 → 0.4483
Diffusion, seed 17	0.8000 → 0.8103	0.2343 → 0.9370	0.4537 → 0.4602
Diffusion, seed 27	0.9091 → 0.9091	0.2342 → 0.9368	0.5088 → 0.5088

A Claim-Ledger Map

The claim ledger `vons_tech_report_v1_1_claims.csv` (the 492 rows of the v1.0.1 ledger unchanged—258 supplied v2-* rows, 227 TR2-* rows added in v0.2 for derived values, documented statuses and disclosures, 7 TR2-* source-code binding rows added in v1.0.1—plus PF-* rows added in v1.1 for every post-freeze value, with the handoff-stated `source_path` and `source_sha256`, the label “POST-FREEZE MEASURED (Codex-verified handoff; not independently rerun by Claude Science)” and the verification note “HANDOFF: value and hash as supplied by Codex on 2026-09-25; file not accessed”) is a private review file that records, for every row, `source_path`, `source_sha256`, `json_pointer`, `operation`, `value_json`, `scope`, `label`, `v1_0_location` and, new in v1.1, `v1_1_location`, `v1_1_section` and `verification`. Table 21 groups the rows by the report location that carries them and names the evidence files each group rests on. Paths under the packet’s private review directory are shown as “internal review record (not public)”.

Table 21. Traceability map from report locations to claim-ledger rows and evidence-packet sources.

Location	Ledger rows	Evidence sources
Table 7	V2-001–004	<code>artifacts/shared-v1/manifest.json</code>
Section 5.2	V2-005–009, TR2-028–034	<code>reports/shared-v1/parity.json</code>
Table 4	V2-010–030	<code>reports/quality-summary-v2.json</code>
Table 5	V2-031–051	<code>reports/quality-summary-v2.json</code>
Table 6	V2-052–060, TR2-065–075	<code>reports/pilot-direct-test-calibrated-v2.json</code>
Table 9	V2-061–078	<code>reports/browser-direct-wasm-trae-summary.json</code> ; <code>reports/browser-direct-webgpu-trae-summary.json</code>
Table 10	V2-079–258	<code>reports/mind2web/evaluation-replay-v1/summary.json</code>
Section 1; Table 15	TR2-001–007, TR2-224	<code>packet.json</code> ; <code>publication-evidence-v2.zip</code> ; <code>reports/publication-claims-v2.csv</code>
Table 7; Section 5.1	TR2-008–027	<code>artifacts/shared-v1/manifest.json</code> ; <code>artifacts/provenance/browser-direct/bundle-manifest-v1.json</code> ; <code>artifacts/provenance/browser-diffusion/bundle-manifest-v1.json</code>
Section 4.1	TR2-035–047, TR2-080–081, TR2-215–219	<code>reports/quality-summary-v2.json</code> ; <code>data/generated/pilot-calibration.manifest.json</code> ; <code>data/generated/pilot-test.jsonl</code> ; <code>artifacts/pilot/model-manifest.json</code>
Section 6.2	TR2-048	<code>reports/quality-summary-v2.json</code>
Section 4.2; Section 4.3	TR2-049–064	<code>reports/quality-summary-v2.json</code>

Location	Ledger rows	Evidence sources
Table 2; Section 7.2	TR2-076–077	reports/pilot-direct-test-calibrated-v2.json
Section 4.5	TR2-078, TR2-211–212	reports/pilot-direct-test-calibrated-v2.json; reports/quality-summary-v2.json
Section 4.4; Section 11	TR2-079	reports/pilot-direct-test-calibrated-v2.json
Table 8; Section 5.3	TR2-082–104, TR2-220–223	reports/onnx-shape-decomposition.json
Table 9; Section 5.4	TR2-105–132	reports/browser-direct-wasm-trae-summary.json; reports/browser-direct-webgpu-trae-summary.json; reports/browser-direct-wasm-merged.json
Section 5.4 (parity not established)	TR2-133–134	internal review record (not public)
Table 10; Section 6.1	TR2-135–153	reports/mind2web/evaluation-replay-v1/ summary.json
Section 6.1 (acquisition)	TR2-154–163	reports/mind2web/data-provenance-v1.json
Section 6.1 (protocol)	TR2-164–165	reports/mind2web/prediction-run-v1.json
Table 11; Section 7.1	TR2-166–192	reports/local-model-alignment-v1.json
Table 2; Section 2; Section 11	TR2-193–198	reports/orchestration-checkpoint-v2.json
Section 5.4; Section 11	TR2-199	reports/orchestration-checkpoint-v2.json
Section 7.2	TR2-200–204, TR2-207	internal review record (not public); docs/ MODEL_ATTRIBUTION.md
Section 12; Section 13	TR2-205	pyproject.toml
Section 2	TR2-206, TR2-209, TR2-232–233	docs/KASI_INTEGRATION.md; docs/ BENCHMARK_PROTOCOL.md; vons/contract.py; tools/ verify_browser_parity.py
Table 3	TR2-208	docs/COMPLETION_PLAN.md
Section 13	TR2-210	drafting-session disclosure
Section 4.3	TR2-213–214	reports/quality-summary-v2.json
Section 9.3	TR2-225–226	tools/predict_mind2web.py; tools/ import_mind2web.py
Table 2	TR2-227	publication-evidence-v2.zip
Section 4.1 (metric definitions)	TR2-228–230	vons/evaluation.py
Section 2; Section 3.3; Section 11	TR2-231	sdk/typescript/src/onnx.ts
Section 1; Table 15; Section 11	TR2-234	packet.json
Section 1; Section 8; Table 17	PF-001	handoff: reports/goal-v2/evidence-summary.json (POST-FREEZE (HANDOFF))
Table 12; Section 8.1; Table 13	PF-002–017	handoff: reports/onnx-shape-decomposition- v2.json; reports/shared-v1/parity-v2-readonly.json (POST-FREEZE (HANDOFF))
Table 12; Section 8.2; Table 14	PF-018–050	handoff: pilot-v2 data statement; vons.calibration/v3 fit; calibrated Direct test report (POST-FREEZE (HANDOFF))
Table 12; Section 8.3	PF-051–075	handoff: Diffusion export manifest; ONNX graph; synthetic test report (POST-FREEZE (HANDOFF))
Table 12; Section 8.4	PF-076–100	handoff: Mind2Web v2 prediction index; replay summary; frozen baseline percentage (derived) (POST-FREEZE (HANDOFF))
Section 8.5; Table 2; Section 2	PF-101–107	handoff: implementer checks (POST-FREEZE (HANDOFF))

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